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Biophysical Insights into Repeat-RNA Clustering from Coarse-Grained Molecular Dynamics

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Abstract

Repeat-expansion RNAs can self-assemble through multivalent RNA–RNA interactions, forming intracellular condensates that disrupt RNA processing and contribute to neurological and neuromuscular disease. Yet, how intrinsic RNA properties determine condensation propensity and the morphology of the resulting assemblies remains poorly understood.

This thesis used coarse-grained molecular dynamics simulations to isolate how saturable Watson–Crick attraction, chain stiffness, repeat length, sequence patterning, and mixture composition control CAG-repeat RNA clustering. RNA was represented at single-nucleotide resolution using a minimal model with chain connectivity, bending stiffness, and valency-limited complementary interactions.

In multi-chain systems, clustering depended non-monotonically on base-pairing strength. Weak attraction produced predominantly dispersed molecules, whereas intermediate attraction distributed contacts across many molecular partners and stabilized large connected networks. At high attraction, the total number of contacts approached the finite bonding capacity, but contacts became increasingly intramolecular or concentrated between fewer molecular pairs. Large-scale connectivity was consequently lost in favor of persistent, locally saturated monomers and small oligomers. Moderate chain stiffness suppressed intramolecular looping and redirected bonding toward intermolecular partners, while greater stiffness restricted the geometries available for broadly connected networks.

Increasing CAG-repeat length lowered the interaction strength required for clustering at fixed nucleotide concentration. By contrast, disrupting the periodic CAG arrangement had only limited effects: repeat and composition-matched scrambled sequences produced broadly similar cluster structures, with significant differences in mean pairwise connectivity detected only for the shortest chains. In mixed CAG10/CAG47 systems, the higher-valency long chains provided a connected scaffold that recruited short chains into shared, heterotypically bonded assemblies.

Overall, the results show how interaction strength, molecular geometry, and valency combine to generate non-monotonic and composition-dependent repeat-RNA assembly.

List of Abbreviations

CG coarse-grained

FIRE fast inertial relaxation engine

HOOMD-blue Highly Optimized Object-oriented Many-particle Dynamics

LLPS liquid–liquid phase separation

MD molecular dynamics

MFE minimum free energy

PBC periodic boundary conditions

RBP RNA-binding protein

WC Watson-Crick

List of Symbols

E_{angle} Harmonic angle potential energy controlling chain stiffness

E_{att} Watson–Crick attractive well amplitude (free parameter of the model)

E_{bond} Harmonic backbone bond potential energy

E_{pot} Total potential energy of the system

E_{wc} Pairing energy contribution from the Watson–Crick interaction

E_r Repulsive barrier amplitude of the soft-core potential

G Molecular contact graph; nodes are molecules and edges connect molecule pairs meeting the contact threshold

L Cubic simulation box length, $L = V^{1/3}$

N_E Number of qualifying molecule-pair edges in the molecular contact graph

N_{A} Avogadro's constant

$N_{\text{MFE}}^{\text{ref}}$ Total number of base pairs in the ViennaRNA minimum free energy reference structure

N_{MFE} Number of formed pairs that also appear in the MFE reference structure

N_{bp} Total number of Watson–Crick contacts after filtering

N_{extra} Number of formed pairs that do not appear in the MFE reference structure

N_{inter} Total number of intermolecular Watson–Crick contacts after filtering

N_{intra} Total number of intramolecular Watson–Crick contacts after filtering

N_{mol} Number of RNA chains in the system

N_{p} Total number of particles in the system

N_{steps} Total number of integration steps

N_{uniq} Number of unique contacts observed within a rolling window

R_g Radius of gyration of a chain

$U(r)$ Pairwise soft-core Watson–Crick base-pairing potential

$U_a(r)$ Attractive component of the soft-core potential

$U_r(r)$ Repulsive component of the soft-core potential

V Simulation box volume

Δt Integration timestep

$\Omega(n)$ Density-dependent switching function suppressing attraction beyond the valency threshold

γ Langevin friction coefficient

$\langle k \rangle$ Mean graph degree, defined as the average number of qualifying molecular partners per molecule, $\langle k \rangle = 2N_E/N_{\mathrm{mol}}$

KE Total kinetic energy of the system

$\phi(r)$ Smooth distance kernel weighting neighbor contributions to the local density

ρ Number density, $\rho = N_{\mathrm{p}}/V$

θ_0 Equilibrium angle of the harmonic bead–bead–bead potential

b_0 Equilibrium bond length of the harmonic bond potential

c Molar strand concentration, $c = N_{\mathrm{mol}}/(N_{\mathrm{A}}\,V)$

f_{inter} Fraction of contacts that are intermolecular

f_{intra} Fraction of contacts that are intramolecular, $f_{\mathrm{intra}} = 1 - f_{\mathrm{inter}}$

kT Thermal energy (Boltzmann constant times temperature) in reduced simulation units

k_{bbb} Harmonic angle stiffness constant of the backbone bead–bead–bead angle

k_b Harmonic bond force constant

l_{RNA} Length of a single RNA chain in beads (one bead per nucleotide)

n_{cut} Valency threshold for the density-dependent switching function

n_{min} Minimum number of intermolecular base-pair contacts required for two molecules to be considered associated

p_c Particle fraction of a cluster, $p_c = s_{\mathrm{c}}\,l_{\mathrm{RNA}}/N_{\mathrm{p}}$

r_ℓ Inner cutoff of the density kernel

r_a Attraction cutoff distance

r_e Equilibrium bonding distance, $r_e = (r_r + r_a)/2$

r_r Repulsion cutoff distance

r_u Outer cutoff of the density kernel

s_{c} Cluster size, defined as the number of molecules in a connected component of G

w_{ij} Molecule-pair edge weight, defined as the number of intermolecular Watson–Crick contacts between molecules i and j

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1 Introduction

Biomolecular condensates are membraneless compartments that concentrate proteins and nucleic acids, thereby localizing conditions for biochemical reactions in the cell [1]. These structures support diverse cellular functions in both the nucleus and cytoplasm. Examples include the nucleolus, which drives ribosome biogenesis and sequesters misfolded proteins [2], stress granules, which transiently store stalled mRNAs and RNA-binding proteins (RBPs) during cellular stress [3], and processing bodies (P-bodies), which regulate mRNA translation and decay [4].

A key mechanism for the formation of biomolecular condensates is liquid–liquid phase separation (LLPS), a process in which biomolecules such as proteins and nucleic acids demix from the surrounding solution to form a dense condensate phase that coexists with a dilute phase [1, 5]. A major driver of LLPS is molecular multivalency: the ability of a single macromolecule to engage in multiple interactions on distinct binding sites [6, 7]. These non-covalent interactions include hydrogen bonding as well as electrostatic and aromatic contacts, and their formation and stability depend on temperature and ionic strength, which together set the saturation concentration for demixing [1].

Early studies of biomolecular condensation largely focused on proteins, particularly RBPs containing intrinsically disordered or low-complexity regions [8, 9]. Lacking stable tertiary structure, these regions expose aromatic and charged residues that engage in the dense multivalent contacts required for demixing, with RBPs such as FUS, TDP-43, and hnRNPA1 as canonical examples [10, 11, 1]. Dysregulation of LLPS has been directly implicated in neurodegenerative diseases. In particular, aberrant phase separation of FUS and TDP-43 is linked to amyotrophic lateral sclerosis and frontotemporal dementia [10, 12].

RNA has been shown to regulate the phase behavior of RBPs. In vitro studies of prion-like RBPs demonstrate that RNA modulates LLPS in a concentration-dependent manner, with low RNA-to-protein ratios promoting phase separation and high ratios suppressing it [13]. High concentrations of nuclear RNA help maintain the solubility of many RBPs by buffering against excessive condensation [13]. Loss of this buffering, for example following protein mislocalization to the relatively RNA-poor cytoplasm, can promote phase separation [13]. Recent work further suggests that this buffering effect depends not only on RNA concentration but also on RNA length [14]. RNA plays a central regulatory role in these transitions, with disrupted RNA buffering promoting excessive phase separation and the subsequent formation of solid-like, pathogenic aggregates [13].

Protein-free phase separation driven by RNA-RNA interactions has been demonstrated experimentally. Homopolymeric RNAs form protein-free condensates with sequence-dependent physical properties in vitro [15] and can selectively partition other RNAs and peptides into the

condensed phase [16]. Cellular mRNAs exhibit protein-free self-assembly in vitro under stress-like conditions [15]. Stress-granule-associated RNAs are biased toward longer transcripts, consistent with increased multivalency, suggesting that RNA length and composition contribute to stress granule partitioning [17].

Repeat-expansion RNAs have received substantial attention because they are directly linked to neurological and neuromuscular disease through RNA gain-of-function mechanisms, including nuclear foci formation, sequestration of RBPs and disruption of RNA processing [18, 19, 20, 21]. These transcripts contain abnormally long tracts of simple tandem repeats, including CAG, CUG and CGG, generated when microsatellite sequences expand beyond disease-associated thresholds [22, 23]. Expanded repeat transcripts can also contribute to pathology through aberrant protein production, either by repeat-associated non-AUG translation or by splicing events that place the repeat within an in-frame open reading frame [24, 25].

Experimental studies have identified repeat length and intermolecular base pairing as important determinants of repeat-RNA assembly. CAG and CUG repeat RNAs form spherical condensates in vitro, with experiments and computational models indicating the onset of condensation near 30 repeats [20, 26]. This threshold is not universal, however: much shorter CAG constructs containing only six or seven repeats can assemble when duplex formation produces 3' sticky ends that support intermolecular crosslinking [27]. Repeat-RNA condensates also span different material states. CAG and CUG RNAs initially form liquid-like spherical droplets in vitro but rapidly lose internal mobility and develop gel-like properties, consistent with the progressive formation of intermolecular RNA networks [20]. Gel-like CAG-repeat foci have also been observed in the cytoplasm, indicating that this behavior is not restricted to purified in vitro systems [28].

Although experimental techniques have characterized the macroscopic behavior of phase separating repeat RNAs, they provide limited mechanistic insight into how RNA properties drive assembly and determine cluster morphology. This thesis addresses this gap using coarse-grained (CG) molecular dynamics (MD) simulations of a minimal, protein-free RNA model. Rather than aiming to develop a new high-resolution representation of RNA, the objective is to determine how much of the observed assembly behavior can be recovered with a small set of parameters and geometric features: saturable Watson-Crick attraction, chain connectivity, and bending stiffness. Within this framework, base-pairing strength, RNA length, stiffness, and sequence patterning are varied independently to examine their effects on clustering propensity. Beyond cluster size, these questions are analyzed in terms of intermolecular contact networks, resolving differences in connectivity and network organization.

2 Overview of existing RNA coarse-grained models

Existing CG RNA models span a wide range of molecular resolution and computational cost. The present work does not aim to introduce a more detailed RNA representation, but instead uses a minimal model in which individual physical parameters can be varied independently. Representative models are summarized in Table 1 to place the level of coarse-graining used in this thesis in context.

Model	**Resolution**	**Key physical features**	**Primary application**
SIS [26]	1 site/nt	Sequence-specific Watson-Crick pairing with many-body, valency-limited interactions	Repeat-RNA folding and phase separation
CALVADOS-RNA [29]	2 beads/nt	Flexible disordered RNA with effective RNA-RNA and RNA-protein interactions	RNA-protein interactions and biomolecular condensates
RNA2PS [30]	2 beads/nt	Backbone electrostatics, base pairing and stacking, with local sequence-context-dependent interactions	RNA structure, duplex thermodynamics and phase separation
iConRNA [31]	6-7 beads/nt	Base pairing, stacking and local RNA geometry with explicit Mg^{2+} ions	RNA folding and condensation
Martini 3 RNA [32]	Multiple beads/nt	General coarse-grained nonbonded interactions, bonded RNA geometry and explicit solvent and ions	Large RNA and RNA-protein systems
oxRNA [33]	1 rigid body/nt	Anisotropic base pairing, stacking and cross-stacking with explicit nucleotide orientation	RNA folding, hybridization and structural assembly

Table 1: Coarse-grained RNA models spanning different levels of molecular resolution and physical detail.

3 Theoretical background

This chapter establishes the physical framework used to interpret RNA phase separation and assembly. It begins with general thermodynamic principles before considering configurational entropy, reentrant transitions, and the polymer properties that shape RNA association.

3.1 General principles of phase separation

Phase separation is a density transition in which a homogeneous mixture demixes into coexisting phases of different compositions [34]. In LLPS, these are liquid phases, typically a macromolecule-rich dense phase (the condensate) and a macromolecule-poor dilute phase [35]. Crucially, phase separation and condensation refer to systems for which coexistence between dense and dilute phases has been established, whereas clustering refers more generally to the formation of finite connected molecular assemblies [35].

Thermodynamically, phase separation occurs when separating a mixture lowers its total free energy relative to a single-phase homogeneous solution [34, 36]. In the simple lattice model of N sites, the spontaneity of mixing is set by the free energy of mixing per lattice site:

$$\frac{\Delta F_{\mathrm{mix}}}{N k_{\mathrm{B}} T} = \underbrace{x_A \ln x_A + x_B \ln x_B}_{-\frac{\Delta S_{\mathrm{mix}}}{N k_{\mathrm{B}}}} + \underbrace{\chi x_A x_B}_{\frac{\Delta H_{\mathrm{mix}}}{N k_{\mathrm{B}} T}}, \tag{1}$$

where x_A and x_B are the mole fractions of the two components and χ is the dimensionless exchange parameter quantifying the relative strength of A-B interactions [36]. The entropic contribution to the free energy of mixing is negative and favors mixing, reflecting the number of ways the two components can be arranged across the lattice. The mixing enthalpy term captures the energetic cost or benefit of A-B contacts and can favor either mixing or demixing depending on the sign of χ [36]. For small molecules occupying single lattice sites, mixing is entropically favorable ($\Delta S_{\mathrm{mix}} \gg 0$), as it maximizes the number of accessible spatial configurations the system can adopt [36].

In polymer systems, by contrast, the entropy of mixing per unit volume is much smaller [34]. Flory and Huggins formalized this with a mean-field model with the free energy of mixing per lattice site:

$$\frac{\Delta F_{\mathrm{mix}}}{N k_{\mathrm{B}} T} = \frac{\phi_A}{N_A} \ln \phi_A + \frac{\phi_B}{N_B} \ln \phi_B + \chi \phi_A \phi_B, \tag{2}$$

where ϕ_A, ϕ_B are volume fractions, N_A, N_B the degrees of polymerization of each species, and χ the Flory interaction parameter representing the enthalpic cost of A-B contacts relative to the average of A-A and B-B contacts [37, 38]. For a polymer, $N_A \gg 1$, so the entropic term

$\phi_A/N_A \ln \phi_A$ is suppressed by a factor of N_A. This follows directly from chain connectivity: because monomers are covalently linked, a chain cannot sample the lattice as N_A independent particles but only as a single connected object, which sharply reduces its translational entropy relative to that of N_A free unlinked monomers [36, 34]. Consequently, mixing long chains produces a substantially smaller increase in the number of accessible microstates than mixing the same number of independently moving monomers [36, 34].

With the reduced entropic gain of mixing for polymers, even weak attractive interactions between chains can promote demixing. Biological polymers achieve these interchain contacts through molecular multivalency, i.e. the presence of multiple interaction sites on a single molecule [1, 39]. Individually weak, monovalent interactions may therefore act cooperatively, stabilizing intermolecular association and promoting assembly [1, 35, 39].

However, a crucial consideration not captured by Flory-Huggins theory is the effect of intrachain associations and their competition with intermolecular bonding. This competition is especially relevant for RNA where the same binding sites may be used to form secondary structures or to cluster neighboring strands via intermolecular contacts [40, 7, 41].

Semenov and Rubinstein improved the general Flory-Huggins theory by considering intrachain bonding in the sticker-spacer model, in which a polymer is represented as a chain of associating sites, or stickers, separated by non-associating spacers [42]. Sticker-sticker bonds are pairwise and saturable, and the total bonding capacity of the system is the sum of all sticker-sticker interactions [42]. Intrachain loops are considered between adjacent stickers and included as a correction to the mean-field free energy as they use up the finite bonding capacity without promoting clustering [42]. This approach captures the reduction in intermolecular bonding capacity caused by local self-association, but does not account for the configurational multiplicity of alternative inter- and intramolecular bonding arrangements or its effect on the resulting state of assembly.

3.2 Configurational entropy and reentrant transitions

Configurational multiplicity is a relevant consideration for repeat RNAs. Thermodynamic studies of CNG repeat RNAs show that these sequences can populate competing duplex, hairpin and shifted-register conformations [43]. This structural heterogeneity arises from the repeat architecture itself: because the same short sequence unit is repeated many times, similar local base-pairing contacts can be redistributed among alternative folds rather than specifying a single dominant RNA structure [20, 26]. Shifted folds such as slipped hairpins can expose repeat units that form contacts with other RNA molecules, providing a structural route from single-chain folds to dimers [44]. In longer repeat RNAs, the same principle can extend to higher-order clustering when multiple repeat segments mediate contacts across several strands [20, 41]. Folded monomers, dimers, oligomers and larger clusters therefore represent compet-

ing arrangements of distributing a finite number of base-pairing sites among different numbers of RNA strands.

Building on the associative sticker-spacer model introduced above, Kimchi et al. developed an analytical model that relates this competition to the configurational multiplicity of different multimeric states [41]. As in the Semenov-Rubinstein model, each strand is represented by n saturable stickers which can individually form at most one bond. However, rather than treating intrachain association only through an averaged local-loop correction, the Kimchi model distinguishes structures containing different numbers of strands and accounts for the number of intra- and intermolecular bonding arrangements available to each state. A connected structure containing m strands may contain both intramolecular and intermolecular bonds, provided that all strands are connected through intermolecular contacts. The equilibrium prevalence of m-strand structures is set by their partition function [41]:

$$Z_m = e^{-\beta(m-1)\Delta F} \sum_{N_\mathrm{b}} g(n,m,N_\mathrm{b})\, e^{-\beta F N_\mathrm{b}}. \tag{3}$$

The first exponential factor represents the multimerization penalty, including the loss of translational and orientational freedom when each additional RNA strand is incorporated into an assembly. The third factor accounts for the free energy gain associated with forming sticker-sticker bonds, set by the number of bonds N_b and the bonding strength F [41]. A key additional contribution is the multiplicity factor $g(n,m,N_\mathrm{b})$, which counts the number of distinct connected bonding arrangements containing N_b bonds among m strands, each carrying n stickers [41]. Two structures with the same N_b carry different statistical weight if one admits more bonding patterns, and this multiplicity grows fastest for large multimers, which can distribute a fixed number of bonds across far more intra- and intermolecular combinations than the same strands kept as monomers or dimers. Phase separation therefore depends not on the maximum number of bonds alone but on the balance between bond energy, the cost of multimerization, and configurational entropy [41].

This balance produces three regimes as sticker strength increases. When interactions are weak, bonds are too unstable to form at all, so the system remains dominated by monomers. At intermediate strength, forming bonds is favorable but complete saturation is not required. Leaving some stickers unpaired costs some bond energy but preserves many possible bonding arrangements. This configurational multiplicity grows steeply with multimer size. Its entropic gain can therefore outweigh both the cost of the unsatisfied stickers and the cost of recruiting additional strands, so aggregation is maximal in this partially bonded regime [41]. At high strength, bond energy dominates and nearly all stickers must be paired. Small folded monomers or dimers can already saturate their stickers, so larger aggregates gain no further bonding energy while still paying the cost of association. Full saturation also restricts the number of accessible bonding

patterns. The configurational advantage of large multimers is lost, and the system reverts to small, fully paired structures [41].

Together, these regimes, from unassociated monomers, to large clusters, back to saturated small structures, constitute a reentrant transition: increasing interaction strength does not monotonically promote assembly, but suppresses it at both extremes for microscopically distinct reasons and maximizes it only at intermediate strength, where configurational entropy of partial bonding is greatest [41]. Where this reentrant regime falls depends on both concentration and molecular architecture, including sticker number and linker length [41]. While reentrant behavior can arise biologically through various mechanisms, such as excess RNA saturating protein binding sites in RNA–protein condensates [13], in this thesis it is considered only in the context of RNA–RNA interactions.

3.3 Polymer properties governing RNA assembly

To conceptualize RNA within the associative polymer framework, its specific molecular architecture must be considered. RNA is a linear polymer of nucleotides, consisting of a negatively charged ribose-phosphate backbone and aromatic, hydrogen-bonding nucleobases [40]. This structure yields a diverse repertoire of potential interaction sites enabling canonical and non-canonical base pairing, π-stacking, and electrostatic interactions with ions and proteins [40, 7].

Persistence length

Because these interaction sites are covalently linked along the backbone, their relative spatial positioning is governed by the chain's conformational properties. RNA is a semiflexible polymer: over short contour distances, the backbone resists bending and preserves its trajectory, whereas over longer distances, it flexibly explores conformational space [34]. This rigidity is quantified by the persistence length, l_p, defined as the characteristic contour distance over which tangent vectors along the chain remain correlated [34]. A larger l_p denotes a stiffer chain, which imposes stricter geometric constraints on the relative proximity of interaction sites.

RNA persistence length emerges from intrinsic and electrostatic contributions. Intrinsic stiffness arises from local backbone geometry and base-stacking interactions, both of which depend on nucleotide chemistry and sequence. For example, weakly stacking poly(U) RNA has a shorter contour length than poly(dT), which can be attributed to differences in sugar-pucker geometry and residual base stacking [45]. Electrostatic repulsion between negatively charged phosphate groups provides an additional contribution that favors more extended chain conformations. Cations, most notably divalent Mg^{2+}, screen this repulsion, reducing the electrostatic contribution to stiffness and promoting chain compaction [45]. RNA persistence length is therefore not fixed, but varies with nucleotide identity, local structural context, and ionic conditions.

In the context of RNA phase separation, the persistence length has been shown to affect condensate morphology [46]. In coarse-grained simulations of poly(U)–polycation mixtures, flexible RNA chains assembled into compact, spherical droplets, whereas rigid chains formed elongated network-like structures [46]. More generally, chain stiffness constrains the relative positions and orientations that interaction sites can adopt. It can therefore alter the probability of forming intramolecular loops, the accessibility of intermolecular contacts, and the number of distinct bonding arrangements available to an RNA molecule. Persistence length may consequently shift the balance between intra- and intermolecular association, although this relationship has not yet been systematically characterized [41, 30].

Length

RNA length shapes condensation behavior across both cellular and reconstituted systems. mRNAs localized in cellular stress granules are strongly biased toward longer sequences [47, 17]. These results are mirrored in vitro, where protein-free RNA self-assembly is biased toward longer transcripts that substantially overlap with the stress-granule transcriptome observed in vivo [17]. Condensation of homopolymeric RNAs such as poly(A) is also length dependent; increasing chain length shifts phase boundaries to lower Mg^{2+} concentrations, whereas the short $poly(A)_{10}$ does not separate even at very high Mg^{2+} concentrations [48].

Repeat-expansion RNAs show a similar threshold behavior. In vitro studies of repeat CAG and CUG RNAs identified a length threshold at around 30 repeats (90 nucleotides) at which condensates begin to form [20]. This threshold is supported by CG simulations which identify a critical length between 20 and 30 repeats, also showing that longer chains form more intermolecular base pairs and higher viscosity condensates [26]. Further simulation work supports that decreasing RNA length to 20 repeats weakens phase separation propensity while 10 repeats do not phase separate at all [31]. Recent work found that condensates formed by CAG and CUG repeats became increasingly stable once the RNA exceeded roughly 30 repeats, which the authors ascribed to the greater intermolecular connectivity of longer chains [30].

The enhanced clustering of longer RNA chains can be attributed to both reduced translational entropy and increased molecular valency. According to Flory-Huggins theory, the polymer contribution to the mixing entropy scales as $1/N$, such that longer chains experience a smaller entropic advantage from remaining dispersed at fixed nucleotide concentration [37, 38]. In addition, longer chains contain more potential interaction sites and can bridge a greater number of molecular partners. This higher effective valency promotes intermolecular network formation and lowers the connectivity threshold for the formation of large clusters.

Sequence composition and patterning

Sequence composition has multifaceted effects on RNA's propensity for condensation. In simple terms, composition determines the stability of base-pairing interactions. Homopolymeric RNAs exhibit an experimentally observed stability hierarchy of poly(G) > poly(A) > poly(C) > poly(U), which broadly parallels their condensation propensities [15]. Poly(G) is stabilized by strong stacking and G-quadruplex formation, whereas poly(U) is comparatively weakly structured and conformationally mobile [15, 49]. For heterogeneous sequences, interaction energies cannot be assigned from nucleotide composition alone. Base-pair stability depends on the identities and arrangement of neighboring nucleotides, while the formation of loops, bulges and other structural motifs introduces additional sequence- and geometry-dependent free-energy contributions, as represented in nearest-neighbor models of RNA folding [50].

The repetitive architecture of CAG RNA, with regularly spaced segments capable of forming consecutive G-C base pairs, creates a dense pattern of complementary interaction sites [51]. The repeated CAG pattern also allows strand slippage between alternative pairing registers, generating partially paired hairpins with unpaired terminal overhangs that can initiate intermolecular association [44, 43]. In vitro studies support a role for sequence periodicity, as composition-matched scrambled RNAs remained soluble under conditions in which $(\mathrm{CAG})_{47}$ formed micrometre-scale gels [20]. However, scrambled CAG sequences can also phase-separate, with scrambling weakening condensation for $(\mathrm{CAG})_{31}$ but enhancing it for $(\mathrm{CAG})_{20}$, indicating that the effect of patterning may depend on chain length and experimental conditions [52]. Coarse-grained simulations likewise found that a composition-matched scrambled sequence could undergo phase separation, but with a lower propensity than $(\mathrm{CAG})_{47}$, forming smaller and less densely packed droplets at a higher saturation concentration [26]. This difference was attributed to the reduced probability of aligning similarly patterned regions on scrambled chains for intermolecular base pairing [26]. Apart from these somewhat contradictory findings, the mechanistic role of sequence periodicity in repeat RNA phase separation has not been clarified.

Mixtures

In cells, condensates may contain many RNAs of varying lengths. Given the substantial effect of length on PS propensity, this raises the question of how RNAs of different lengths partition between the dilute and dense phases in mixed systems. Coarse-grained simulations of protein and mixed-length RNA condensates show that longer RNA concentrates in the core to maximize protein-RNA cross-linking, while shorter RNA accumulates at the interface as a biomolecular surfactant to reduce interfacial free energy, enabling this dual mechanism to stabilize condensates [53]. Whether these principles extend to protein-free condensates composed of RNAs of different lengths remains largely unexplored.

4 Aims

RNA assembly emerges from competition between intramolecular folding and intermolecular association for the same finite set of base-pairing sites. How this competition is affected by chain stiffness and binding strength, and how it interacts with RNA length, sequence patterning, and mixture composition, remain poorly understood. The overarching aim of this thesis is to disentangle these contributions using a minimal coarse-grained model in which each property can be varied independently. In particular, the thesis tests the reentrant dependence of clustering on binding strength predicted by Kimchi et al. [41] and examines how RNA properties shape the connectivity and composition of the resulting assemblies.

As a prerequisite for studying RNA clustering, the model is evaluated for its ability to reproduce the predicted stem-loop structure of an isolated RNA chain.

Single-chain characterization

At the single-chain level, the model is first validated against a reference RNA hairpin to establish how stiffness and binding strength reshape chain conformation and base-pairing dynamics in isolation, and under which combination the simulated ensemble recovers the predicted secondary structure and chain compactness.

Building on this single-chain characterization, three research questions address collective RNA assembly:

RQ1 - Reentrant transition

The sticker-spacer framework of Kimchi et al. [41] predicts a non-monotonic dependence of aggregation on sticker binding strength, with clustering emerging at intermediate values and suppressed in both the weak- and strong-binding regimes. Does the present model reproduce this reentrant behavior, and where along the binding strength axis does it occur? How does chain stiffness shift the transition boundaries?

RQ2 - Length and patterning effects

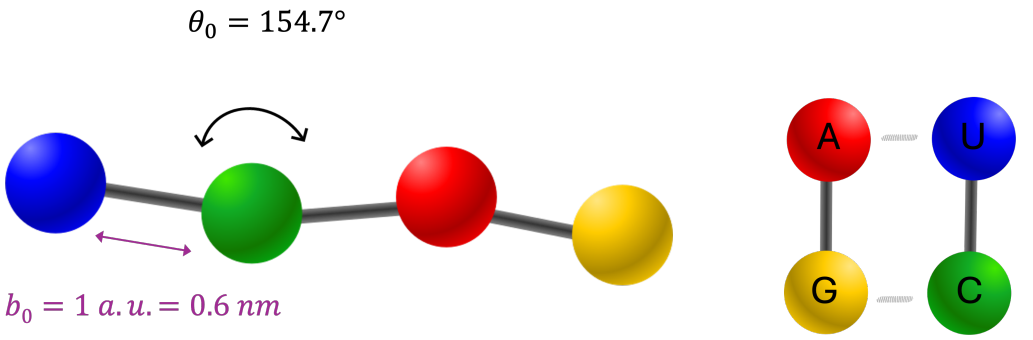
At constant total particle count, does increasing CAG-repeat length promote clustering, and is the periodic CAG register itself required? Each repeat is compared against a composition-preserving scrambled control to disentangle the effects of length and patterning.

RQ3 - Mixed-length systems

Do $(\mathrm{CAG})_{10} + (\mathrm{CAG})_{47}$ mixtures form mixed, species-enriched, or species-segregated clusters? How does clustering compare to homogeneous systems of the two species?

5 Methods

Coarse-grained molecular dynamics simulations were performed using a Langevin integrator in Highly Optimized Object-oriented Many-particle Dynamics (HOOMD-blue) version 5.4.0 [54]. RNA is modeled at single-nucleotide resolution with interactions for chain connectivity, bending stiffness, and canonical Watson-Crick pairing, as detailed in the following sections. The model architecture is summarized schematically in Figure 1.



(a) Coarse-grained RNA representation.

(b) Watson-Crick pairing.

Figure 1: Schematic of the coarse-grained RNA model. **(a)** Each nucleotide is represented by a single bead. Adjacent beads are connected by harmonic bonds with equilibrium length b_0, while θ_0 sets the preferred bending angle between consecutive particles. **(b)** Complementary A-U and G-C pairs receive the attractive component of the valency-limited Watson-Crick interaction.

5.1 Coarse-grained representation of RNA

RNA is a polymer of nucleotides, each consisting of a ribose sugar, a phosphate group, and a nucleobase. In the coarse-grained model, each RNA nucleotide is represented as a single spherical bead carrying the identity of one of the four nucleobases: adenine (A), guanine (G), uracil (U), or cytosine (C). The ribose-phosphate backbone is omitted, and the model operates under the assumption of fully screened electrostatic interactions.

Chain connectivity and stiffness

Adjacent beads along the chain are connected by harmonic bonds of the form

$$U(b) = \tfrac{1}{2} k_b (b - b_0)^2, \tag{4}$$

with equilibrium bond length $b_0 = 1$ simulation unit, corresponding to approximately 0.6 nm, and $k_b = 100$. The bond length is consistent with the inter-phosphate distance in RNA, where the C3'-endo sugar pucker enforces A-form helical geometry with consecutive phosphates separated by 5.9 Å[55].

Chain stiffness is modeled with a harmonic angle potential acting on each consecutive triplet of beads [B-B-B],

$$U(\theta) = \tfrac{1}{2}\, k_{\mathrm{bbb}}\, (\theta - \theta_0)^2, \tag{5}$$

with an equilibrium angle of $\theta_0 = 154.7^\circ$ (2.70 rad). This value was set to the mean backbone angle observed in all-atom MD simulations of RNA double-helix structures [56].

Calibration of chain stiffness to RNA persistence length

In the coarse-grained model, the combined effects of local backbone geometry, base stacking, and screened electrostatic interactions are represented by the harmonic angle stiffness k_{bbb}. This parameter was calibrated against the experimentally measured persistence length of single-stranded poly(U) RNA.

For a discrete worm-like chain, bond-vector orientational correlations decay exponentially with contour separation s:

$$\langle \cos \alpha(s) \rangle = \exp\left(-\frac{s b_0}{l_p} \right), \tag{6}$$

where $\alpha(s)$ is the angle between bond vectors separated by s bonds, b_0 is the equilibrium bond length, and l_p is the persistence length [34]. For rU_{40}, experimental measurements give a high-ionic-strength limiting persistence length of approximately 10Å, increasing to approximately 25Å at low ionic strength [45].

To calibrate k_{bbb} independently of secondary-structure formation, single-chain simulations were performed with a homopolymeric RNA sequence. Chains of length $l_{\mathrm{RNA}} = 1000$ were simulated across k_{bbb} values ranging from 0 to 200. Bond-vector correlations were computed for bond separations up to $s = 50$ and fitted to the exponential decay above. Interpolation of the resulting relationship between l_p and k_{bbb} gave $k_{\mathrm{bbb}} = 2.5$ for the target persistence length of approximately 10Å (Figure 2).

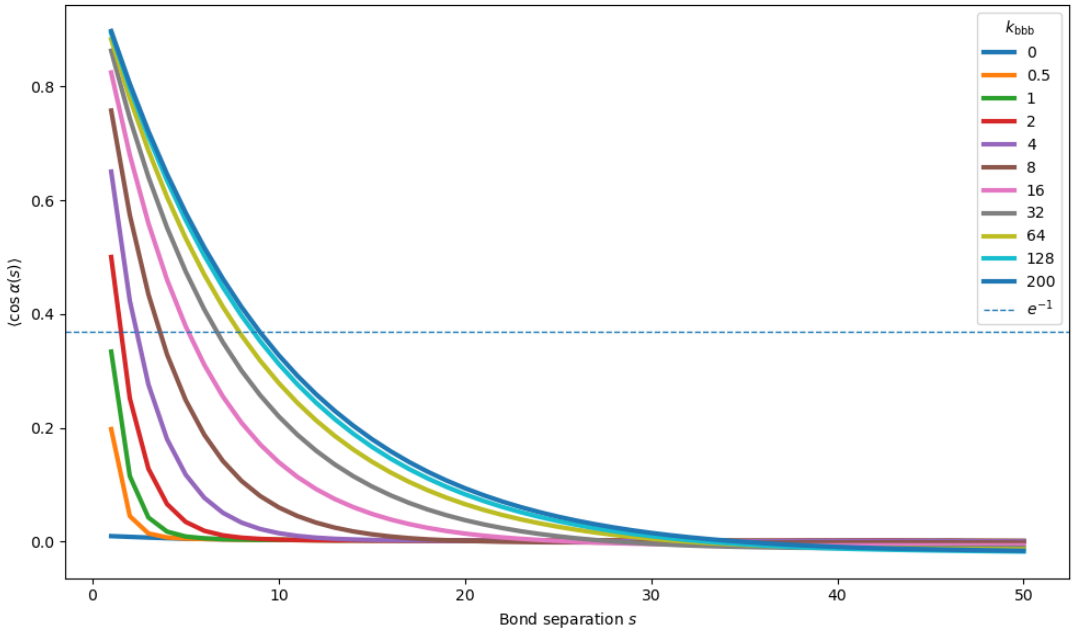


Figure 2: Bond-vector correlation $\langle \cos \alpha(s) \rangle$ as a function of bond separation s up to $s = 50$, for varying k_{bbb}. The dashed line marks e^{-1}; at its intersection with each curve, the corresponding bond separation $s_{1/e}$ is related to the persistence length by $l_p = s_{1/e} b_0$.

The value $k_{\text{bbb}} = 2.5$ was subsequently used as the primary stiffness value in production simulations, with other k_{bbb} values also explored to examine the effect of increased chain rigidity.

5.2 Watson-Crick base-pairing potential

The nonbonded WC base-pairing interactions are modeled using a many-body soft-core potential $U(r)$, split into a repulsive and an attractive component.

$$U(r) = U_r(r) + U_a(r) \tag{7}$$

The repulsive part $U_r(r)$ acts at short range, preventing particle overlap:

$$U_r(r) = E_r \left(1 - \left(\frac{r}{r_r}\right)^2\right)^2 \Theta(r_r - r) \tag{8}$$

The attractive part $U_a(r)$ defines the bonding well centered at $r_e = (r_r + r_a)/2$:

$$U_a(r) = -E_{\text{att}} \left(1 - \left(\frac{r - r_e}{\frac{r_a - r_r}{2}}\right)^2\right)^2 \Theta(r - r_r)\,\Theta(r_a - r) \tag{9}$$

where E_r and E_{att} set the energy scales of the repulsive and attractive components, respectively,

r_r is the repulsion cutoff, r_a is the attraction cutoff, and $\Theta(\cdot)$ is the Heaviside step function ensuring each term acts only within its specified distance range. The negative sign of the attractive contribution is included explicitly in $U_a(r)$, so E_{att} is reported throughout as a positive quantity.

Density-dependent valency limit

To approximate single-bond valency, such that each base preferentially forms a WC contact with only one partner at a time, the attractive component is modulated by a density-dependent switching function:

$$n_i = \sum_{j \neq i} \phi(r_{ij}) \tag{10}$$

where $\phi(r)$ is a smooth kernel that weighs the contribution of each eligible complementary particle by distance:

$$\phi(r) = \begin{cases} 1 & r < r_\ell \\ \left(1 - \left(\dfrac{r - r_\ell}{r_u - r_\ell}\right)^2\right)^2 & r_\ell \leq r \leq r_u \\ 0 & r > r_u \end{cases} \tag{11}$$

Here r_ℓ and r_u are the inner and outer cutoffs of the density shell. Neighbors closer than r_ℓ contribute fully, those beyond r_u are ignored, and those in between contribute partially.

The full density-corrected potential for a pair (i, j) is then:

$$U(r_{ij}) = U_r(r_{ij}) + \Omega(n_i)\,\Omega(n_j)\,U_a(r_{ij}) \tag{12}$$

where the switching function $\Omega(n)$ suppresses attraction once a particle's neighbor count exceeds a threshold n_{cut}:

$$\Omega(n) = \begin{cases} 1 & n < n_{\text{cut}} \\ \left(1 - (n - n_{\text{cut}})^2\right)^2 & n_{\text{cut}} \leq n \leq n_{\text{cut}} + 1 \\ 0 & n > n_{\text{cut}} + 1 \end{cases} \tag{13}$$

The product $\Omega(n_i)\,\Omega(n_j)$ suppresses attraction if either particle has a sufficiently high local neighbor density, creating an effective valency limitation by discouraging multiple close neighbors.

Soft-core potential parameters

The potential parameters used in all simulations are summarized in Table 2. The repulsive amplitude $E_r = 20$ was chosen to be sufficiently large to prevent particle overlap. The attractive amplitude E_{att} is a free parameter of the model varied across a range of values to map out clustering transitions. In the analytical sticker-spacer model by Kimchi et al. [41], sticker binding strength depends on sequence and can be estimated from nucleic-acid thermodynamics, whereas E_{att} is an independent reduced interaction-strength parameter. Sequence and interaction strength are therefore varied independently, and no direct mapping between E_{att} and a sequence-specific binding free energy is assumed.

The valency threshold $n_{\text{cut}} = 1.061$ was set just above 1 to produce effective monovalent WC bonding per base, while the small tolerance above 1 allows for bond switching transitions.

Parameter	Value	Description
E_r	20	Repulsive barrier amplitude
E_{att}	varied	Attractive amplitude
r_r	0.25	Repulsion cutoff distance
r_a	0.75	Attraction cutoff distance
r_e	0.50	Attractive well minimum
r_ℓ	0.48	Density inner cutoff
r_u	0.807	Density outer cutoff
n_{cut}	1.061	Valency threshold

Table 2: Parameters of the WC soft-core potential used in all simulations. All distances are given in simulation units.

The base-pairing potential was developed in the Goloborodko lab and implemented in C++ as a custom HOOMD-blue [54] plugin. The three components of the potential evaluated with these parameters are shown below (Figure 3).

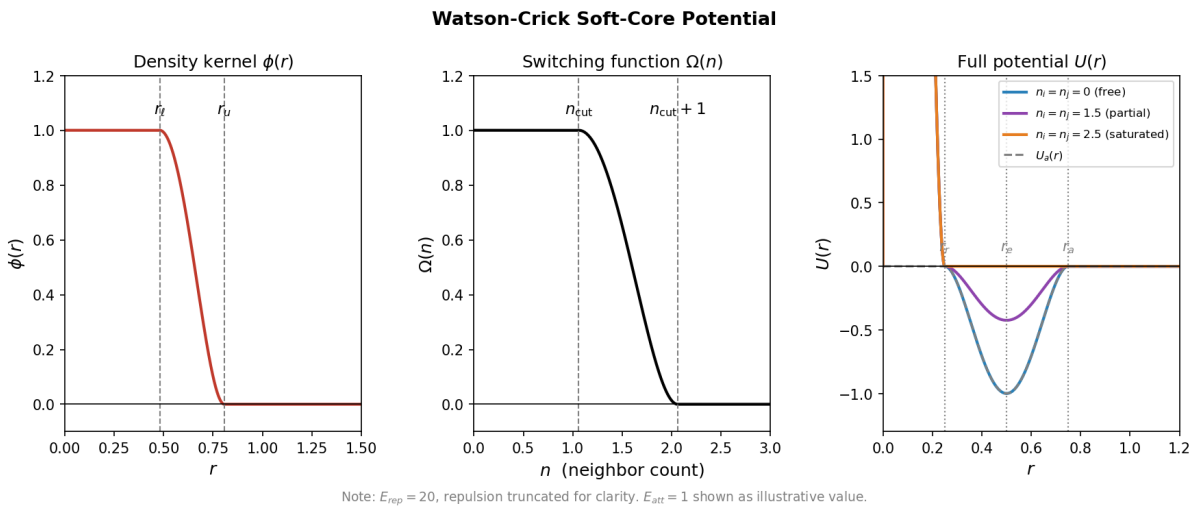


Figure 3: WC soft-core potential components. *Left:* Density kernel $\phi(r)$ with inner and outer cutoffs r_ℓ and r_u marked. *Center:* Switching function $\Omega(n)$ as a function of neighbor count n, with valency threshold n_{cut} marked. *Right:* Full potential $U(r)$ for three values of neighbor density, showing how the attractive well is progressively suppressed as a particle becomes bonded.

Base-pairing specificity and interaction exclusions

Only complementary A-U and G-C pairs received the attractive contribution, with both pair types assigned the same attraction strength. Other nucleotide combinations interacted only through the repulsive term. Although the interaction matrix could be extended to include distinct A-U and G-C strengths or additional pairs such as G-U wobble interactions, this thesis uses a minimal sticker-spacer description of saturable complementary binding.

Nonbonded interactions were excluded between directly bonded nucleotides and between nucleotides separated by two or three bonds along the same chain. In HOOMD-blue, this was implemented using bond, 1-3, and 1-4 exclusions in the neighbor list, removing intramolecular pairs with sequence separation $|i-j| < 4$ from the Watson-Crick potential. This prevents local neighbors from forming pairing contacts and precludes hairpin loops containing fewer than three intervening nucleotides, consistent with the minimum loop size commonly used in RNA secondary-structure models [57].

5.3 Molecular dynamics simulations

Simulations were run in HOOMD-blue version 5.4.0 [54] in a cubic simulation box with periodic boundary conditions. Initial particle positions were set with a self-avoiding random walk centered at the origin, then relaxed by fast inertial relaxation engine (FIRE) energy minimization. Particle momenta were initialized from a Maxwell-Boltzmann distribution at $kT = 1$ and the system was integrated with Langevin dynamics using friction coefficient $\gamma = 0.1$. The integration timestep $\Delta t = 0.01$ and friction coefficient $\gamma = 0.1$ were selected for optimal integration

stability and equilibration speed.

The cubic box length is determined from the target particle number density,

$$\rho = \frac{N_\mathrm{p}}{V}, \tag{14}$$

where N_p is the total number of particles and V is the box volume in simulation units. The resulting box length is $L = V^{1/3}$. Periodic boundary conditions (PBC) were applied in all simulations to mimic a bulk solution and avoid surface effects.

To relate the simulation density ρ to biological molar concentration, the box volume is converted to liters using the length mapping of $b_0 = 0.6$ nm, making the volume of one cubic simulation unit, $(0.6\,\mathrm{nm})^3 = 2.16 \times 10^{-25}\,\mathrm{L}$. The molar strand concentration is then:

$$c = \frac{N_\mathrm{mol}}{N_\mathrm{A} \cdot V} \quad [\mathrm{mol\,L^{-1}}] \tag{15}$$

where N_mol is the number of RNA molecules, N_A is Avogadro's constant, and V is the box volume in liters. Molar strand concentration serves only as a physical reference, as all research questions are evaluated at fixed particle number.

Production simulations

All production simulations shared the integration protocol described above: a cubic box with PBC, with the edge length set by the particle number density as $L = (N_\mathrm{p}/\rho)^{1/3}$. After FIRE energy minimization of the initial configurations, the systems were integrated using Langevin dynamics at $kT = 1$, with friction coefficient $\gamma = 0.1$ and timestep $\Delta t = 0.01$. The systems, sequences, and parameter sweeps for each research question are summarized in Table 3.

Single-chain characterization

A single 30-nucleotide hairpin (Table 6) was simulated at a reduced density $\rho = 0.005$, chosen to suppress self-interaction across the periodic boundary. Each trajectory was run for $N_\mathrm{steps} = 3 \times 10^7$ steps. The parameter grid spanned $E_\mathrm{att} \in \{1, 2, 3, 4, 8\}$ and $k_\mathrm{bbb} \in \{0, 2.5, 5\}$, with five replicates per condition. All structural quantities were computed over the final 20% of each trajectory.

RQ1 – Reentrant transition

$N_\mathrm{mol} = 100$ molecules of $(\mathrm{CAG})_{47}$ were simulated at $\rho = 0.01$ (corresponding to a strand concentration of 0.545 mM). The Watson-Crick attraction was tested at $E_\mathrm{att} \in \{1, 2, 4, 6, 8, 12, 16, 32\}$ and the stiffness at $k_\mathrm{bbb} \in \{0, 2.5, 5\}$, with ten replicates per condition. All simulations were

run for $N_{\text{steps}} = 4 \times 10^7$ steps. Cluster and contact metrics were computed over the final 10% of each trajectory.

RQ2 – Length and patterning

$(\text{CAG})_x$ systems with $x \in \{10, 20, 31, 47\}$ were simulated at a nearly fixed total particle count ($N_{\text{p}} \approx 14\,100$) and fixed density $\rho = 0.01$, so that shorter repeats were represented by proportionally more chains (Table 3). Consequently, the molar strand concentrations were 2.56, 1.28, 0.827, and 0.545mM for $(\text{CAG})_{10}$, $(\text{CAG})_{20}$, $(\text{CAG})_{31}$, and $(\text{CAG})_{47}$, respectively. The attraction was restricted to the intermediate clustering regime, $E_{\text{att}} \in \{2, 4, 8\}$, at $k_{\text{bbb}} = 2.5$, with five replicates per condition. Each repeat-length system was run for $N_{\text{steps}} = 4 \times 10^7$ steps. Metrics were computed over the final 10% of each trajectory.

To test the effect of the periodic CAG pattern on clustering, a scrambled control was generated for each repeat length. Each control replicate contained the same number of molecules as the corresponding repeat system, and every molecule retained the same chain length and exact numbers of A, C, and G particles. Only their order along the chain was changed. Each molecule was scrambled independently, so molecules within the same system generally had different sequences. The complete set of scrambled sequences was regenerated for every replicate (Figure 4). To better sample the additional sequence variability in scrambled controls, they were run with n=10 replicates, compared with n=5 for the repeat sequences. Metrics were computed over the final 10% of each trajectory. Each scrambled control system was run for $N_{\text{steps}} = 4 \times 10^7$ steps.

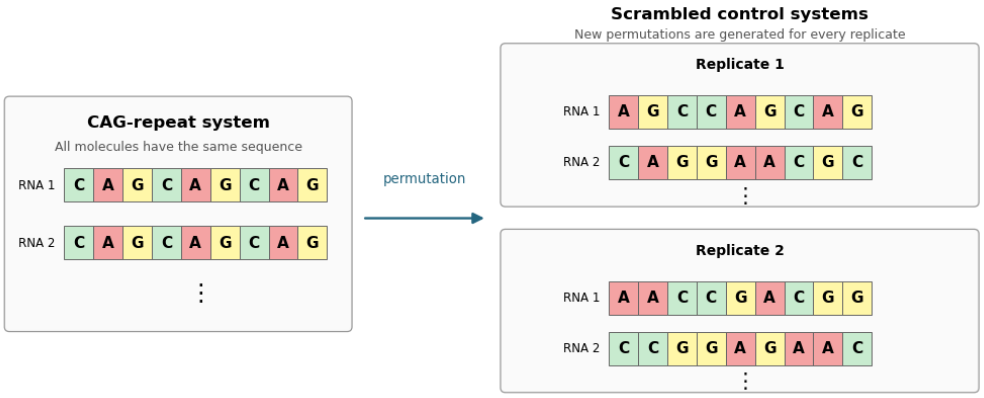


Figure 4: Generation of scrambled controls. Each control comprised an ensemble of independently scrambled molecules with the same chain length and A, C, and G composition as the corresponding CAG-repeat molecules. A new sequence ensemble was generated for every replicate. Shortened sequences are shown for illustration.

RQ3 – Mixed-length systems

A binary mixture of 235 $(\mathrm{CAG})_{10}$ and 50 $(\mathrm{CAG})_{47}$ chains was simulated at $\rho = 0.01$ and $k_{\mathrm{bbb}} = 2.5$. The composition was chosen so that each species contributed an equal number of particles ($235 \times 30 = 50 \times 141 = 7\,050$), matching the total particle count of the single-length systems ($N_{\mathrm{p}} = 14\,100$). This corresponds to a total strand concentration of approximately 1.55 mM, comprising 1.28 mM $(\mathrm{CAG})_{10}$ and 0.273 mM $(\mathrm{CAG})_{47}$. The attraction was swept over $E_{\mathrm{att}} \in \{2, 4, 8\}$, and the mixture was compared against pure $(\mathrm{CAG})_{10}$ and pure $(\mathrm{CAG})_{47}$ reference systems from RQ2 at matched density and stiffness. The mixed systems were simulated for $N_{\mathrm{steps}} = 6 \times 10^7$ steps at 10 replicates per condition. Metrics were computed over the final 5% of each trajectory.

RQ	**System(s)**	N_{mol}	N_{p}	ρ	E_{att}	k_{bbb}
Single-chain	30-nt hairpin	1	30	0.005	1, 2, 3, 4, 8	0, 2.5, 5
RQ1	$(\mathrm{CAG})_{47}$	100	14 100	0.01	1, 2, 4, 6, 8, 12, 16, 32	0, 2.5, 5
RQ2	$(\mathrm{CAG})_{10}$	470	14 100	0.01	2, 4, 8	2.5
	$(\mathrm{CAG})_{20}$	235	14 100	0.01	2, 4, 8	2.5
	$(\mathrm{CAG})_{31}$	152	14 136	0.01	2, 4, 8	2.5
	$(\mathrm{CAG})_{47}$	100	14 100	0.01	2, 4, 8	2.5
RQ3	$(\mathrm{CAG})_{10}$ +$(\mathrm{CAG})_{47}$	235 + 50	14 100	0.01	2, 4, 8	2.5

Table 3: Production simulation systems and parameters.

5.4 Analysis metrics

Logged quantities

Every 10 000 integration steps, particle positions and a set of thermodynamic quantities were saved. Kinetic energy is recorded per particle to allow straightforward monitoring of thermostat stability across systems of different sizes. The total potential energy decomposes additively into three contributions: harmonic bond, harmonic angle, and a Watson-Crick pairing term, so that lower E_{pot} reflects a more energetically favorable state.

Logged quantities are summarized in Table 4.

Quantity	Symbol	Description
Simulation step	$step$	Integration step index
Kinetic energy per particle	KE/N_p	Total kinetic energy normalized by particle count; used to monitor thermostat stability
Total potential energy	E_pot	Sum of all potential energy contributions
WC pairing energy	E_wc	Energy from the soft-core Watson-Crick potential
Bond energy	E_bond	Harmonic backbone bond energy
Angle energy	E_angle	Harmonic angle energy controlling chain stiffness

Table 4: Logged framewise simulation quantities.

Logged energy quantities were used as diagnostics for equilibration.

Base-pair detection

At each saved frame, candidate WC base pairs were identified from particle positions using a neighbor list with cutoff radius $r_\mathrm{max} = 0.7$. The cutoff was chosen slightly below the attraction cutoff $r_a = 0.75$ to exclude near-cutoff encounters from being classified as base-pair contacts.

The neighbor list was further filtered through three criteria to retain only physically valid bonds.

1. **Complementarity:** Only pairs forming valid WC contacts, adenine-uracil (A-U) or guanine-cytosine (G-C), were retained.
2. **Chain exclusion:** Intramolecular candidate pairs separated by fewer than four positions along the chain were excluded, matching the nonbonded neighbor exclusions in the pairing potential.
3. **Valency:** Where a particle appeared in multiple candidate bonds, only the partner closest to the attractive well minimum r_e was retained. A bond was accepted only when this choice was mutual, ensuring that each particle was assigned to at most one partner.

The total number of detected base-pair contacts in each frame is denoted N_bp. Each contact was classified as either intermolecular, when the two particles belonged to different molecules, or intramolecular, when both particles belonged to the same molecule. Accordingly,

$$N_\mathrm{bp} = N_\mathrm{inter} + N_\mathrm{intra}, \tag{16}$$

where N_inter and N_intra are the numbers of intermolecular and intramolecular contacts, respectively. Their relative contributions were quantified as

$$f_\mathrm{inter} = \frac{N_\mathrm{inter}}{N_\mathrm{bp}}, \qquad f_\mathrm{intra} = \frac{N_\mathrm{intra}}{N_\mathrm{bp}} = 1 - f_\mathrm{inter}. \tag{17}$$

The filtered contact list formed the basis for all subsequent single-molecule, molecular-graph, and cluster analyses.

Single-molecule conformation and secondary structure

Chain compactness was quantified by the radius of gyration, computed per frame as the root-mean-square distance of all beads from the chain center of mass:

$$R_g = \sqrt{\frac{1}{l_{\mathrm{RNA}}} \sum_{i=1}^{l_{\mathrm{RNA}}} (\mathbf{r}_i - \mathbf{r}_{\mathrm{cm}})^2} \tag{18}$$

where l_{RNA} is the number of beads in the chain, $\mathbf{r}_i$ is the position of bead i, and $\mathbf{r}_{\mathrm{cm}} = \frac{1}{l_{\mathrm{RNA}}} \sum_i \mathbf{r}_i$ is the center of mass. A low R_g indicates a compact, folded conformation, whereas a high R_g indicates an extended or swollen chain.

Secondary structure agreement was scored by comparing the detected base-pairs against the ViennaRNA minimum free energy (MFE) structure prediction. Each formed pair was classified as an *MFE pair* (N_{MFE}) if it appears in the reference, and as an *extra pair* (N_{extra}) otherwise. To separate the specificity of the formed contacts from the completeness of the recovered structure, two normalized quantities are reported alongside the raw counts: MFE precision, the fraction of formed pairs that are native, and MFE recall, the fraction of reference pairs that are recovered.

$$\mathrm{precision} = \frac{N_{\mathrm{MFE}}}{N_{\mathrm{MFE}} + N_{\mathrm{extra}}}, \qquad \mathrm{recall} = \frac{N_{\mathrm{MFE}}}{N_{\mathrm{MFE}}^{\mathrm{ref}}}. \tag{19}$$

Multi-molecule systems: molecular contact graph

Intermolecular connectivity was represented at each saved frame by a molecular contact graph

$$G = (V, E), \tag{20}$$

where each node $i \in V$ represents one molecule. An undirected edge $(i, j) \in E$ was drawn between molecules i and j when they shared at least $n_{\min} = 2$ intermolecular base-pair contacts in that frame. This threshold prevents a single transient contact from being counted as an intermolecular association.

Each qualifying edge was assigned a weight,

$$w_{ij} = N_{\mathrm{inter}}^{(i,j)}, \tag{21}$$

where $N_{\text{inter}}^{(i,j)}$ is the number of intermolecular base-pair contacts between molecules i and j. Edge weight therefore measures how extensively a particular pair of molecules is bonded. The mean edge weight is the average of w_{ij} over all edges in a frame.

The total edge count,

$$N_E = |E|, \tag{22}$$

is the number of distinct molecule pairs satisfying the contact threshold and therefore measures the overall extent of intermolecular connectivity. The mean graph degree,

$$\langle k \rangle = \frac{1}{N_{\text{mol}}} \sum_{i \in V} k_i = \frac{2N_E}{N_{\text{mol}}}, \tag{23}$$

gives the mean number of distinct qualifying molecular partners per molecule, where k_i is the degree of molecule i. Molecules without any qualifying partners remain isolated nodes with degree zero.

Statistical significance

Differences in mean molecule-pair edge weight between periodic CAG and scrambled controls were assessed using two-sided permutation tests. For each repeat length and E_{att}, the late-time mean edge weight was calculated separately for each replicate, and the repeat/scrambled group labels were permuted to test the difference between group means. The resulting p-values from the 12 comparisons (4 repeat lengths $\times$ 3 attraction strengths) were adjusted for multiple testing using the Holm correction. Statistical significance was defined as adjusted $p < 0.05$.

Cluster classification

Connected components of G define clusters. The size of a cluster is defined either by its number of molecules, denoted s_{c}, or by the fraction of all system particles it contains, denoted p_c. Each cluster was assigned to one of seven size categories (Table 5). For comparisons between systems with the same chain length, clusters were classified by s_{c}. For comparisons between systems containing different chain lengths at matched total particle count, clusters were classified by p_c, while monomers were retained as a separate category.

The reported distributions therefore give the fraction of molecules or particles in each category and sum to one.

Bin	s_c
Monomer	1
Dimer	2
Trimer	3
4-9	$4 \leq s_c < 10$
10-19	$10 \leq s_c < 20$
20-49	$20 \leq s_c < 50$
≥ 50	$s_c \geq 50$

(a) Same-length comparison: bins by raw molecule count s_c.

Bin	Definition
Monomer	$s_c = 1$
$< 2\%$	$p_c \in (0, 2\%)$
2-$< 3\%$	$p_c \in [2\%, 3\%)$
3-$< 10\%$	$p_c \in [3\%, 10\%)$
10-$< 20\%$	$p_c \in [10\%, 20\%)$
20-$< 50\%$	$p_c \in [20\%, 50\%)$
$\geq 50\%$	$p_c \in [50\%, 100\%]$

(b) Cross-length comparison: monomers ($s_c = 1$) are assigned separately; multimers ($s_c > 1$) are binned by particle fraction p_c.

Table 5: Cluster size categories under the two classification schemes.

Ranked cluster sizes

To resolve the largest assemblies more directly, multimeric clusters were ranked independently in each frame by the number of particles they contained. The three largest clusters were reported separately, while all remaining multimeric clusters were combined into an *other multimers* category and unbound molecules into a *monomers* category. The reported contribution of each category is the fraction of total system particles it contains. As clusters were ranked independently in each frame, a given rank does not track the identity of one persistent cluster over time.

Cluster metrics in mixed-length systems

In mixed-length systems of $(\mathrm{CAG})_{10}$ and $(\mathrm{CAG})_{47}$ types, the same molecular contact graph G was used, with each node additionally labeled by molecule type.

For each ranked cluster category, the material contributed by each molecule type was calculated from the number of particles belonging to that type. This decomposition shows how much each chain length contributes to the material contained in the largest clusters, the remaining multimers, and the monomer population.

Intermolecular base-pair contacts within the largest cluster were classified as $(\mathrm{CAG})_{10}$-$(\mathrm{CAG})_{10}$, $(\mathrm{CAG})_{10}$-$(\mathrm{CAG})_{47}$, or $(\mathrm{CAG})_{47}$-$(\mathrm{CAG})_{47}$. For each contact type, both the mean number of contacts and its fraction of all intermolecular contacts in the largest cluster were calculated.

Contact recurrence

The total base-pair count N_{bp} and total edge count N_E record how many base pairs and molecule pairs exist at a given frame, but give no information about their stability. A system may maintain the same number of contacts while being locked into a persistent structure or constantly

exchanging partners. To capture this, contact recurrence was calculated over a rolling window of $w = 20$ most recent frames at both the particle and molecule levels.

For a given definition of contact identity, let N_{uniq} be the number of unique contacts observed at least once within the window, and let $N_c(t)$ be the number of contacts present in frame t. The recurrence score is

$$\text{rec} = 1 - \frac{N_{\text{uniq}}}{\sum_{t=1}^{w} N_c(t)}. \tag{24}$$

A contact present in every frame contributes w occurrences but only one distinct identity. Recurrence is therefore zero when every observed contact is unique and increases when the same contacts repeatedly appear. For a fixed set of contacts present in all frames, the score reaches $1 - 1/w$ and approaches one as the window length increases.

Particle-level recurrence

At the particle level, a contact identity was defined by the pair of beads forming a detected base pair, and $N_c(t) = N_{\text{bp}}(t)$. Applying Equation 24 gives the particle-level recurrence rec_{par}, which measures how frequently the same particle pairs reappear within the rolling window.

Molecule-level recurrence

At the molecule level, a contact identity was defined by a molecule pair connected by an edge in G, and $N_c(t) = N_E(t)$. Applying Equation 24 gives the molecule-level recurrence rec_{mol}. Low values indicate frequent exchange of molecular partners, whereas high values indicate that the same molecule-pair edges persist across the window.

6 Results

6.1 Single-chain characterization

Single-molecule simulations had two goals: to test whether the model reproduces a known RNA secondary structure, and to characterize how the base-pairing strength E_{att} and the angular stiffness k_{bbb} shape the intramolecular conformation in the absence of other chains.

Sequence choice

A 30-nucleotide sequence predicted to fold into a single stable hairpin was selected based on two criteria: a narrow Boltzmann ensemble around a single minimum free energy (MFE) structure, so the target fold is unambiguous, and a short chain length, so the relevant conformations are accessible within reasonable simulation times.

Secondary-structure predictions and equilibrium thermodynamic quantities were obtained with ViennaRNA 2.0 [57]. As an independent estimate of native compactness, R_g^{ref} was computed as the mean over five AlphaFold3 [58] predictions of the same sequence. Reference values are listed in Table 6.

Property	**Value**
Sequence	UUGCCUGACUAACGAUCGUUAGUCAGGCAA
MFE dot-bracket structure	.((((((((((((....)))))))))))).
Ensemble free energy	-20.94 kcal/mol
MFE structure frequency	49.06%
Ensemble diversity	0.66
R_g (AlphaFold3, mean of 5)	1.47 nm

Table 6: Thermodynamic and structural reference values of the 30-nt RNA sequence used for single-chain characterization. The ensemble free energy (-20.94 kcal/mol) gives the total free energy of the Boltzmann ensemble; more negative values indicate a more thermodynamically stable fold. The MFE structure frequency (49.06%) is the Boltzmann probability of the MFE structure within the ensemble, and the ensemble diversity (0.66) is the average base-pair distance between structures in the ensemble, with lower values indicating structural homogeneity.

The MFE frequency of 49.06% indicates that the MFE structure carries nearly half of the Boltzmann probability, while the low ensemble diversity (0.66) indicates that structures within the ensemble differ from one another by less than one base pair on average. Together, these values indicate that the ensemble is dominated by the MFE hairpin and closely related alternative structures.

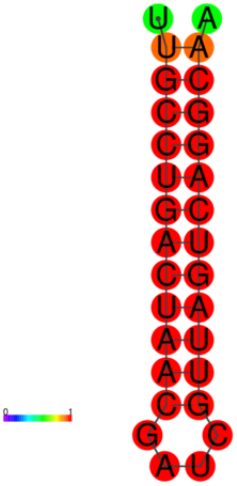
Simulation setup

Simulations were run for $N_{\text{steps}} = 3 \times 10^7$ steps at a reduced density $\rho = 0.005$ to avoid self-interaction across the periodic boundary. The parameter grid spanned $E_{\text{att}} \in \{1, 2, 3, 4, 8\}$ and

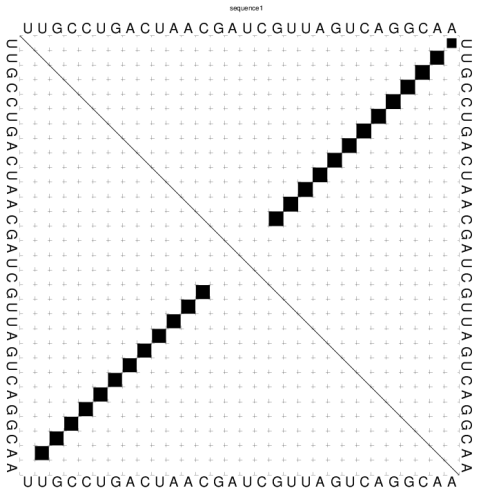
$k_{\mathrm{bbb}} \in \{0, 2.5, 5\}$, with five replicates per condition. All structural quantities were computed on the final 20% of each trajectory.

Secondary structure

Base pairs formed during the simulation were compared against the ViennaRNA MFE pair list. The MFE structure (Figure 5a) consists of a 12-pair stem closed by a 4-nucleotide loop, with one A and one U unpaired at the termini. The base-pair probability dot plot (Figure 5b) shows that the equilibrium partition function places nearly all weight on these 12 pairs, confirming the absence of competing low-energy folds.



(a) MFE structure with per-pair probabilities.



(b) Dot plot: MFE pairs (lower triangle) and partition-function pairing probabilities (upper triangle).

Figure 5: ViennaRNA reference for the chosen sequence. (a) The MFE structure is a 12-pair stem with a 4-nucleotide loop. (b) Equilibrium pairing probabilities concentrate on the MFE pairs.

In each saved frame, the base-base contacts produced by the simulation were sorted into pairs that also appear in the MFE structure (MFE pairs) and pairs that do not (extra pairs). This gives two complementary axes for evaluating the fold: how many MFE pairs are recovered, and how many extra pairs accompany them (Figure 6).

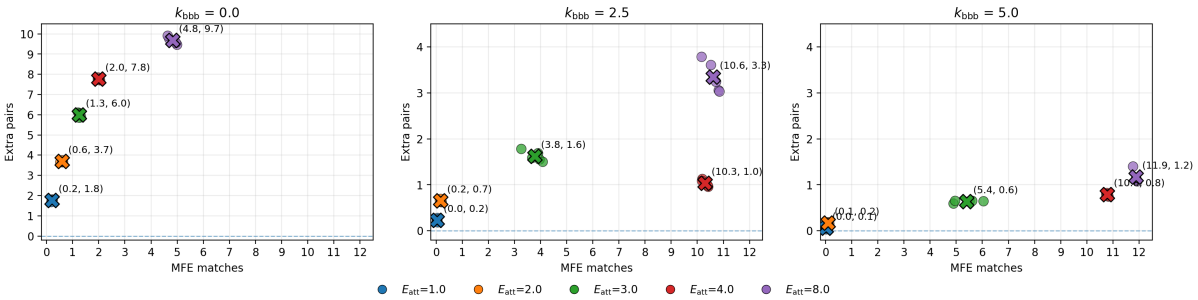


Figure 6: MFE-matching pairs versus extra pairs in the final 20% of each trajectory across three stiffness regimes. Each circle represents one replicate, colored by E_{att}, and each cross marks the mean across five replicates. The annotations beside the crosses report the mean number of MFE-matching pairs first, followed by the mean number of extra pairs.

For fully flexible chains ($k_{bbb} = 0$), both MFE and extra pairs grew with E_{att} but the system never reached the reference fold: at $E_{att} = 8$, ~ 9.6 MFE pairs were accompanied by ~ 4.8 extra pairs, indicating a misfolded ensemble with a partially formed stem and spurious long-range contacts. At the calibrated stiffness $k_{bbb} = 2.5$, $E_{att} = 4$ yielded ~ 11 MFE pairs with only ~ 1 extra pair, i.e. near-quantitative recovery of the MFE structure. At $k_{bbb} = 5$, recovery was preserved at $E_{att} \in \{4, 8\}$, with the stem essentially fully populated and a small extra contribution.

For direct comparison on a common scale, the same results are presented in terms of MFE precision and recall in Figure 7.

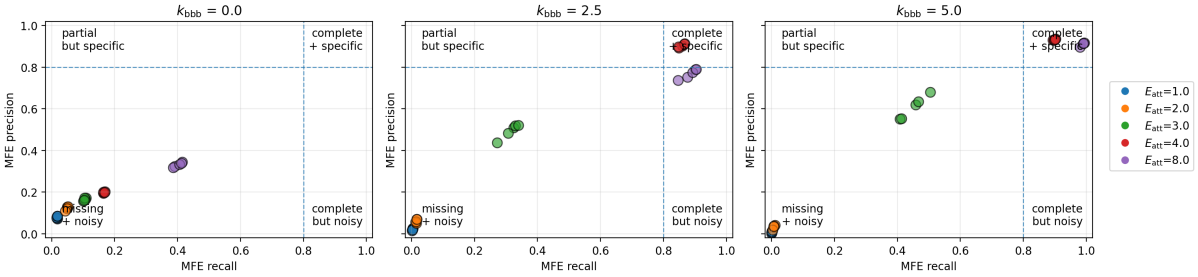


Figure 7: MFE precision-recall quadrants for the final 20% of each trajectory across k_{bbb}. Each point is one replicate, colored by E_{att}. Dashed lines at 0.8 define four diagnostic regimes: *complete + specific* (correct fold), *partial but specific* (correctly registered but incomplete stem), *complete but noisy* (MFE pairs present alongside many extra pairs), and *missing + noisy* (misfolded).

The precision-recall plot exposes a clear interplay between angular stiffness and attraction strength. At $k_{bbb} = 0$, all conditions have low precision and recall: increasing E_{att} produces more MFE-matching pairs, but also many incorrect extra pairs. At $k_{bbb} = 2.5$, the stronger attractions ($E_{att} = 4$ and 8) reach the high-precision, high-recall region, while weaker attractions remain incomplete. A similar trend is observed at $k_{bbb} = 5$, where $E_{att} = 4$ and 8 recover the reference fold most accurately. Overall, increasing k_{bbb} improves precision, whereas increas-

ing E_{att} promotes formation of enough reference pairs and primarily improves recall. Accurate folding requires both sufficient chain stiffness and attraction strength.

Radius of gyration

The global compactness of the chain was compared against the AlphaFold3 reference $R_g^{ref} = 1.47$ nm (Figure 8).

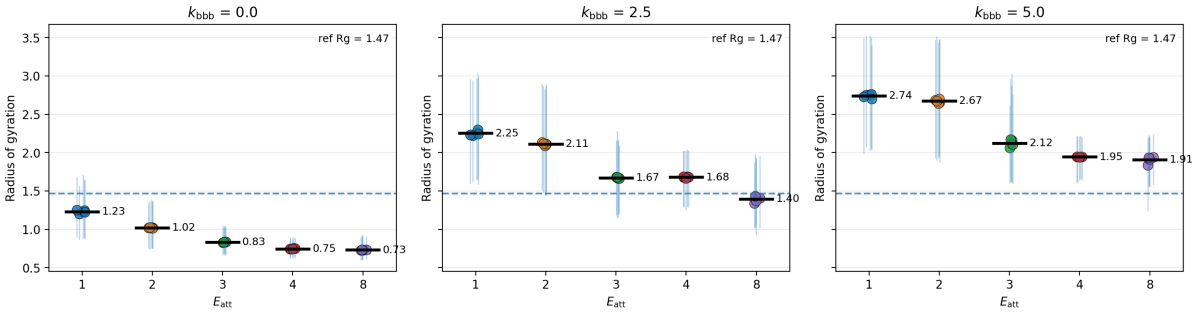


Figure 8: Mean R_g over the final 20% of each trajectory as a function of E_{att} across k_{bbb}. Filled circles show the replicate mean; vertical bars span the 10th-90th percentile range across replicates. The dashed line marks $R_g^{ref} = 1.47$ nm. Numerical labels give the replicate mean per condition.

The radius of gyration reveals the opposing effects of backbone stiffness and attraction strength. At k_{bbb} = 0, the chain is more compact than the AlphaFold3 reference under all conditions, with R_g decreasing from 1.23 to 0.73 nm as E_{att} increases. Together with the low precision in Figure 7, this indicates compact but misregistered structures. At k_{bbb} = 2.5, increasing E_{att} progressively compacts the chain: weak attraction leaves it more extended than the reference, E_{att} = 3 and 4 bring R_g closer to the reference value, and E_{att} = 8 produces slight over-compaction. At k_{bbb} = 5, attraction again promotes compaction, but R_g remains above the reference for every E_{att}, indicating that the high angular stiffness prevents the chain from reaching the reference global compactness.

The condition k_{bbb} = 2.5 and E_{att} = 4 provides the best overall balance between MFE-pair recovery, suppression of extra pairs, and global compactness. Increasing E_{att} to 8 brings R_g closer to the reference but reduces pairing specificity, whereas k_{bbb} = 5 gives high MFE recovery while leaving the chain globally too extended. Thus, some stiffness is necessary to promote specific folding, but at high k_{bbb} the chain becomes less responsive to increasing E_{att}, reducing the extent of attraction-driven compaction.

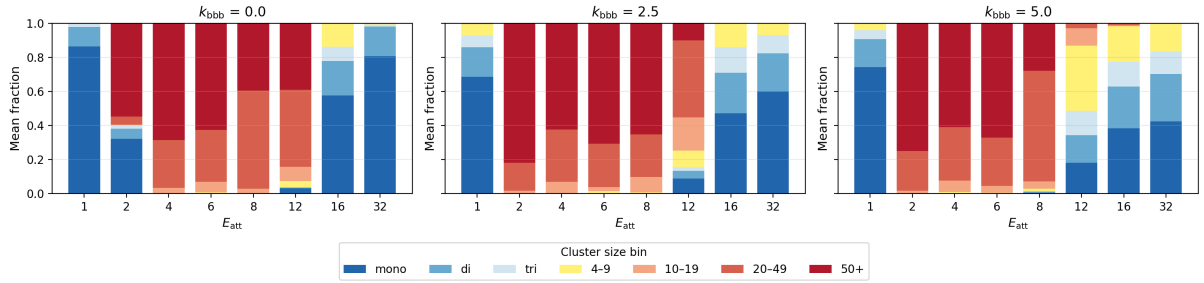
6.2 RQ1: Reentrant transition

Enumeration of nucleic-acid multimers by Kimchi et al. [41] predicts a reentrant response to increasing sticker-sticker binding strength: a monomeric system at weak binding, large clusters at intermediate binding, and a reentrant regime of small saturated multimers at strong binding.

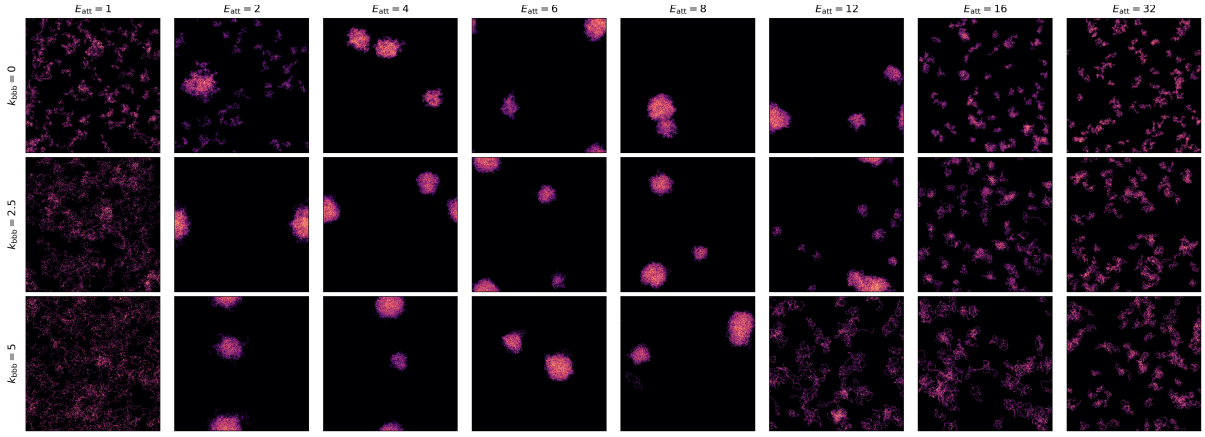
To test whether the coarse-grained model reproduces this response, simulations were performed for $(\mathrm{CAG})_{47}$, a repeat-expanded RNA sequence that readily undergoes phase separation [20]. All simulations contained $N_{\mathrm{mol}} = 100$ molecules at a fixed particle number density of $\rho = 0.01$. The Watson-Crick attraction parameter was varied across $E_{\mathrm{att}} \in \{1, 2, 4, 6, 8, 12, 16, 32\}$. In addition, chain stiffness was varied across three values $k_{\mathrm{bbb}} \in \{0, 2.5, 5\}$, to test its effect on the resulting cluster organization.

Increasing base-base attraction strength produces reentrant clustering

Late-time cluster size composition was used to assess how clustering depends on E_{att} and k_{bbb}. For each condition, cluster-size bin fractions were calculated over the final 10% of each trajectory and averaged across 10 replicates. To complement these quantitative distributions, final-frame 2D projections from a representative replicate were used to visualize the clustering morphology (Figure 9).



(a) Late-time cluster-size composition.



(b) Representative 2D *xy*-projections.

Figure 9: Reentrant clustering across Watson-Crick attraction strengths and chain stiffness values. **(a)** Cluster-size composition over the final 10% of each trajectory. Bars show the mean fraction of molecules in each cluster-size bin across 10 replicates. **(b)** Representative 2D *xy*-projections of the final simulation frame. Columns correspond to increasing E_{att} and rows to k_{bbb}. Molecules are projected onto the *xy*-plane, and structures crossing a panel boundary connect to their periodic images across the opposite boundary. All systems were simulated at $\rho = 0.01$.

Cluster size composition changed non-monotonically with E_{att} (Figure 9a). Increasing E_{att} to intermediate values promoted the formation of large clusters, with the strongest enrichment of the 50+ bin between $E_{att} = 2$ and 8. At higher attraction strengths, $E_{att} = 16$ and 32, large clusters were suppressed again and the distribution shifted back toward monomers and small oligomers. The final-frame projections show the same overall response (Figure 9b): compact assemblies are visible at intermediate attraction strengths, whereas molecules are more broadly distributed in both the weak and strong attraction regimes.

The reentrant response was present at all three stiffness values, but k_{bbb} changed both its onset and the composition of the high-attraction regime. For fully flexible chains, $k_{bbb} = 0$, a substantial monomer population remained at $E_{att} = 2$. At the calibrated stiffness, $k_{bbb} = 2.5$, the same attraction strength was sufficient to place nearly all molecules in the 50+ bin. Increasing the stiffness further to $k_{bbb} = 5$ shifted the system toward smaller oligomers at lower E_{att}, with the 50+ population already absent at $E_{att} = 12$. Chain stiffness therefore changes the range of

attraction strengths over which large clusters are favored, as well as the types of assemblies formed outside this range.

Reentrance follows a shift from inter- to intramolecular bonding

Under the single valency restriction of the soft-core potential used in this thesis, the maximum number of possible contacts is constrained by the finite bonding capacity of the system. For 100 $(CAG)_{47}$ molecules, this capacity is approximately 4700 possible C-G/G-C pairs. Increasing E_{att} therefore does not create an unlimited number of bonds; instead, it increasingly penalizes configurations in which complementary bases remain unsatisfied. At sufficiently large E_{att}, the system is dominated by configurations in which nearly all available base sites are paired.

To determine how the bonding capacity was used, the total contact count was separated into intermolecular and intramolecular contributions (Figure 10).

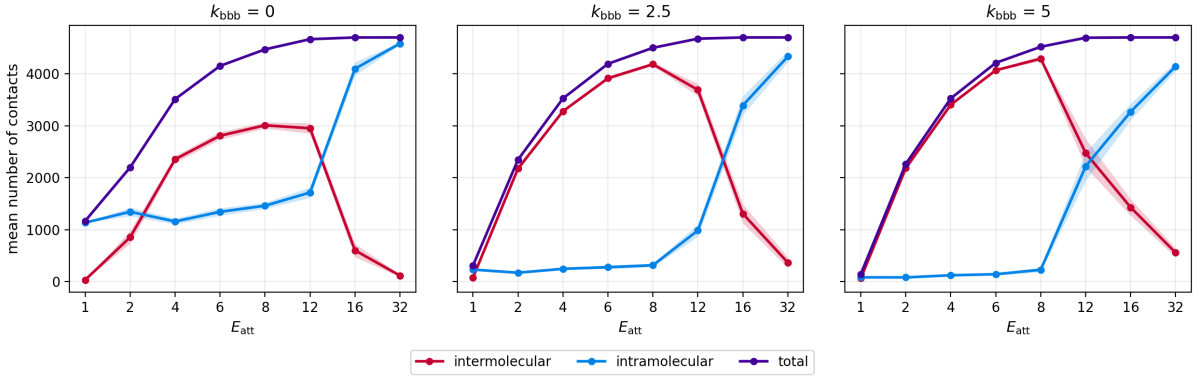


Figure 10: Inter- and intramolecular contacts across E_{att}. Lines show the mean number of intermolecular, intramolecular, and total contacts computed from the final 10% of each trajectory and averaged over 10 replicates. Shaded regions indicate the standard deviation across replicates. Panels show the three tested k_{bbb} values.

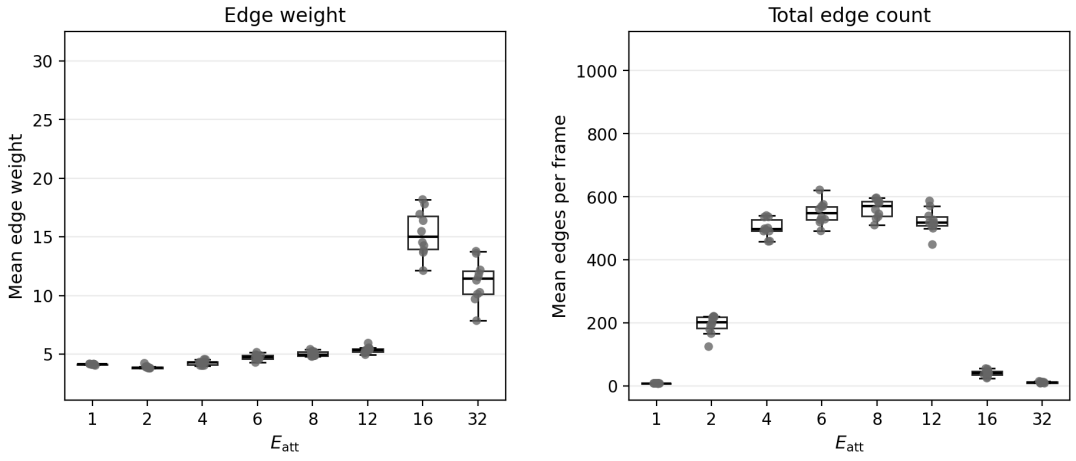
The total number of contacts increased with E_{att} and approached the finite bonding capacity of the system at high attraction strengths (Figure 10). Thus, the reduction in large clusters at high E_{att} did not stem from a loss of base-pair contacts. Instead, the contacts were redistributed. Intermolecular contacts dominated at intermediate attraction strengths, where large clusters were most abundant, whereas intramolecular contacts became dominant at $E_{att} = 16$ and 32, where the cluster-size distribution shifted toward monomers and small oligomers.

Chain stiffness shifts the balance between intra- and intermolecular bonding. Flexible chains ($k_{bbb} = 0$) allocate a substantial fraction of their contacts to intramolecular bonding at low and intermediate attractions (E_{att} =1,2,4,6,8). In contrast, introducing chain stiffness strongly suppresses intramolecular bonding over the same range. At $k_{bbb} = 2.5$, contacts are predominantly intermolecular, while at $k_{bbb} = 5$, intramolecular contacts are almost entirely absent across the low and intermediate attraction regime.

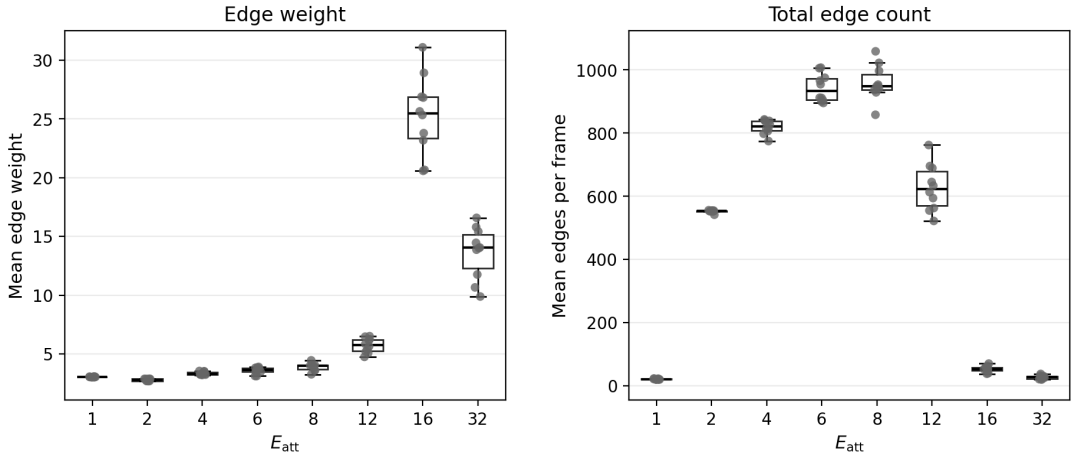
The cluster-size composition discussed in the previous section broadly follows this redistribution between intra- and intermolecular bonding, while also showing that contact type alone does not determine cluster size (Figure 9a). At $E_{\mathrm{att}} = 2$, the substantial intramolecular contact-fraction in flexible chains coincides with a large monomer population. At the same attraction strength and $k_{\mathrm{bbb}} = 2.5$, the fraction of intramolecular bonding is low and accompanied by the incorporation of nearly all molecules into the 50+ bin. However, intramolecular contacts can coexist with large clusters, as observed for fully flexible chains at intermediate attraction strengths. Similarly, for the stiffest chains, the 50+ population disappears at $E_{\mathrm{att}} = 12$ despite the persistence of substantial intermolecular bonding, and the comparatively high intermolecular contribution at $E_{\mathrm{att}} = 16$ and 32 corresponds primarily to smaller oligomers. Cluster size therefore depends both on the balance between intra- and intermolecular bonding and on whether intermolecular contacts connect many molecules into a large network or remain confined to smaller disconnected assemblies.

Similar cluster compositions mask distinct molecule connectivity across the attraction range

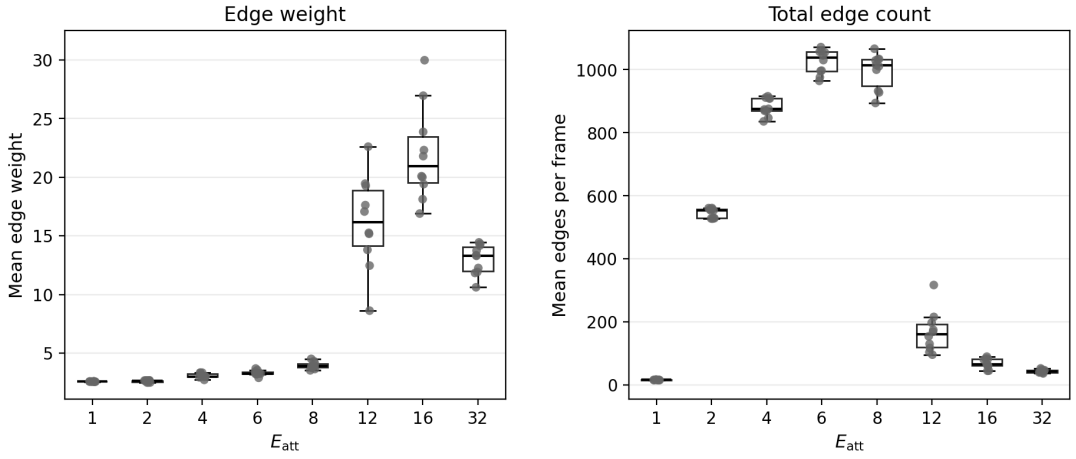
Similar cluster sizes by molecule count can arise from very different underlying connectivity, so cluster-size composition alone provides no information about how densely the molecules within a cluster are connected. To resolve this, properties of the constructed molecule-graph were examined across the full attraction and stiffness range (Figure 11).



(a) $k_{bbb} = 0$



(b) $k_{bbb} = 2.5$



(c) $k_{bbb} = 5$

Figure 11: Molecular contact graph properties across E_{att} and k_{bbb}. For each stiffness, the left plot shows the mean edge weight of the molecular contact graph, while the right plot shows the mean number of total edges per frame. Metrics were calculated over the final 10% of each trajectory at $\rho = 0.01$. Boxplots summarize 10 independent replicates, with individual replicate values shown as points.

At weak attraction, the molecular contact graphs contained few edges for all three stiffness values. Between $E_{att} = 2$ and 8, the total edge count increased strongly with chain stiffness, while the mean edge weight remained low and changed comparatively little. At $E_{att} = 2$, flexible chains formed approximately 200 edges per frame, compared with around 550 in both stiffer systems. The difference remained pronounced from $E_{att} = 4$ to 8, where the two stiffer systems contained approximately 800-1000 edges per frame, compared with 500-560 for flexible chains. Stiffer chains therefore increased the number of distinct molecular connections, reflected in a higher edge count, while mean edge weight remained relatively low. This difference is not apparent from the cluster-size composition, which is dominated by large clusters across this attraction range.

At $E_{att} = 12$, the three stiffness values diverged. The systems with $k_{bbb} = 0$ and 2.5 retained high edge counts with only moderate increases in edge weight. For $k_{bbb} = 5$, the edge count dropped from approximately 1000 to 150, while the mean edge weight increased from around 4 to 16 contacts per pair. Since each edge represents a distinct molecular pair, the contacts were distributed across fewer pairs, with more contacts per pair. This connectivity pattern corresponds to the loss of the 50+ population and its replacement by smaller oligomers.

At $E_{att} = 16$ and 32, the edge count collapsed across all stiffness values, while the remaining edges carried substantially more contacts. The two stiffer systems retained more edges than the flexible system and generally showed higher edge weights, consistent with their larger oligomer populations in the cluster-size composition (Figure 9a). High attraction produced a sparse molecular contact graph in which the remaining edges carried more contacts on average.

The graph analysis reveals a second contribution to the reentrant response. At intermediate attraction, intermolecular contacts are distributed across many molecular pairs, producing the extensive connectivity required for large clusters. At high attraction, multiple contacts accumulate between fewer partners, reducing the number of distinct molecular connections. Extensive intermolecular bonding can therefore coexist with a fragmented network of small oligomers.

Molecule contact recurrence increases with E_{att}

Beyond the count, distribution and weight of molecular connections, their persistence over time reveals whether the assemblies are static or continually rearranging. Molecule-contact recurrence quantifies this, measuring how often bonded molecular partners are retained rather than exchanged across the trajectory (Figure 12).

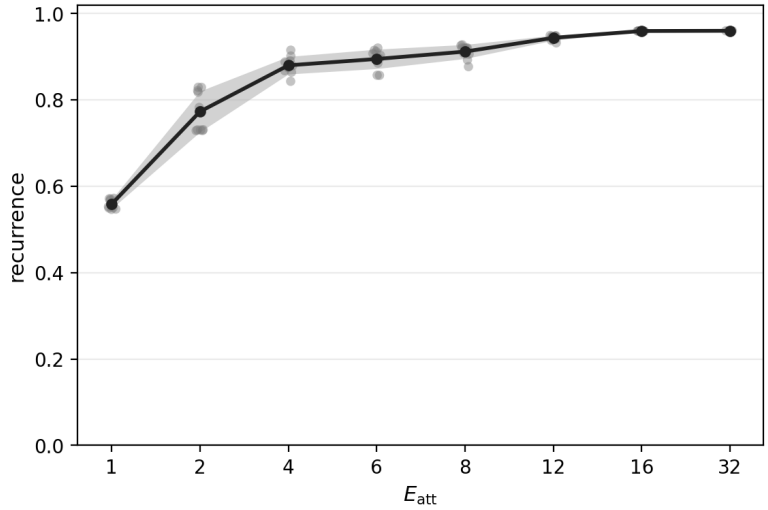


Figure 12: Molecule-contact recurrence across E_{att} for $k_{bbb} = 2.5$. Recurrence was computed from molecule-pair contact identities over the final 10% of each trajectory. Points show individual replicate values, the line shows the mean, and the shaded region indicates the standard deviation across 10 replicates. Simulations were performed at fixed $\rho = 0.01$.

Molecule-contact recurrence increased strongly with E_{att}, rising from highly dynamic contacts at $E_{att} = 1$ to near-persistent molecular partners at intermediate and high attraction strengths. The high recurrence at $E_{att} = 16$ and 32 shows that the smaller clusters formed at high attraction strength are persistent assemblies with long-lived molecular contacts.

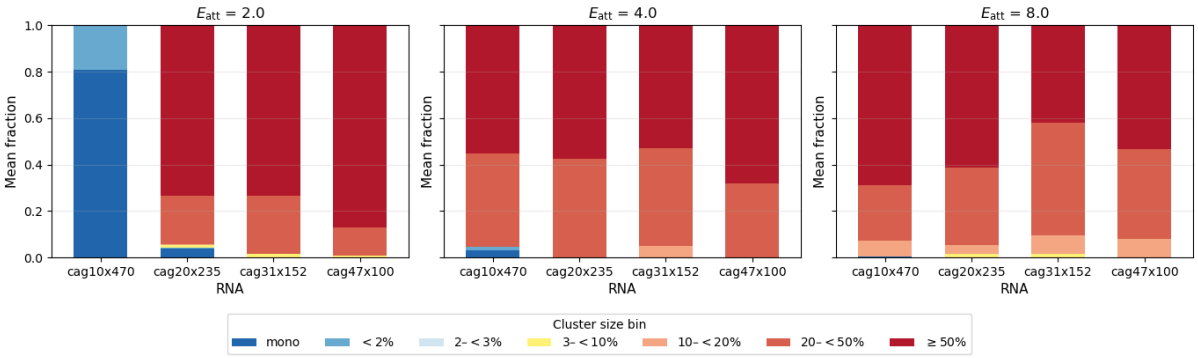
6.3 RQ2: Length and patterning

In vitro studies and computational models have reported that the propensity of repeat RNAs to undergo LLPS depends on length, with phase separation occurring only above a critical repeat-number threshold [20, 26].

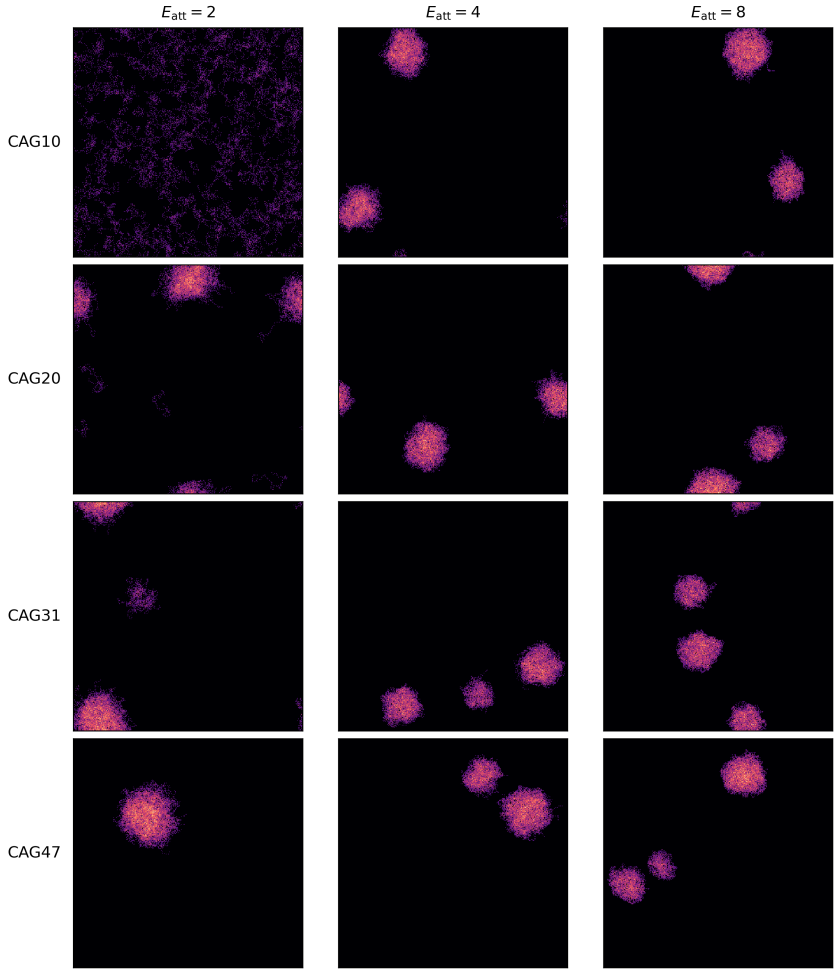
To explore this with the coarse-grained model, simulations were performed for $(\mathrm{CAG})_x$ with $x \in \{10, 20, 31, 47\}$. Chain stiffness was fixed at $k_{bbb} = 2.5$, the value previously calibrated against the experimentally reported persistence length of single-stranded RNA (Figure 2). To isolate repeat length from total nucleotide content, the total number of particles was kept approximately constant at $N_p \approx 14\,100$, while the particle density was fixed at $\rho = 0.01$. Three attraction strengths, $E_{att} \in \{2, 4, 8\}$, were selected from the intermediate regime in which clustering was observed in RQ1.

Longer CAG repeats promote clustering at lower attraction strength

Cluster-size composition and representative 2D density projections were compared across repeat lengths and attraction strengths (Figure 13). Since the total nucleotide concentration was fixed, this comparison tests how distributing the same nucleotide content among more numerous short chains or fewer long chains affects clustering.



(a) Cluster-size composition.



(b) Representative 2D density projections.

Figure 13: Effect of CAG repeat length on RNA clustering at $E_{att} = 2$, 4, and 8. **(a)** Cluster-size composition, with clusters assigned to bins by particle fraction and monomers shown separately. Bar heights indicate the fraction of total particles in each bin over the final 10% of the trajectories. **(b)** RNA density projections for the corresponding repeat lengths and interaction strengths. Systems were simulated at $\rho = 0.01$ and $k_{bbb} = 2.5$.

The strongest repeat-length dependence was observed at $E_{att} = 2$. Approximately 80% of the $(CAG)_{10}$ molecules remained monomeric, with the remaining molecules formed small oligomers. Increasing the repeat length to 20 was sufficient to shift nearly the entire popu-

lation into larger assemblies, with approximately 70% of molecules belonging to the 50+ bin. The contribution of this bin remained high for $(\mathrm{CAG})_{31}$ and increased further for $(\mathrm{CAG})_{47}$. The density projections show the same contrast: the $(\mathrm{CAG})_{10}$ system remained broadly dispersed, whereas the longer repeats formed spatially concentrated assemblies.

At $E_{\mathrm{att}} = 4$, all repeat lengths were dominated by the 20-50% and $\geq 50\%$ bins, and only a small fraction of molecules remained in smaller clusters. Differences between repeat lengths were therefore much weaker than at $E_{\mathrm{att}} = 2$, although $(\mathrm{CAG})_{47}$ showed the largest contribution from the $\geq 50\%$ bin.

At $E_{\mathrm{att}} = 8$, all repeat lengths retained substantial large-cluster populations, but their response relative to $E_{\mathrm{att}} = 4$ differed. The contribution of the $\geq 50\%$ bin increased for $(\mathrm{CAG})_{10}$, whereas it decreased for $(\mathrm{CAG})_{31}$ and $(\mathrm{CAG})_{47}$ as more molecules shifted into the 20-50% bin. This decline may indicate that the reentrant loss of the largest clusters begins at lower attraction strengths for the longer repeats. The limited set of attraction strengths does not allow the position of this transition to be determined precisely.

Repeat length shifts the range of attraction strengths over which the largest clusters are favored. Longer repeats show clustering at $E_{\mathrm{att}} = 2$, while their reduced $\geq 50\%$ contribution at $E_{\mathrm{att}} = 8$ suggests a possible earlier reentrance from the large-cluster regime. In contrast, $(\mathrm{CAG})_{10}$ required stronger attraction to form large clusters, remaining largely monomeric at E_{att} =2.

Particle contact recurrence is low and length-ordered

Particle-level recurrence was used to provide a comparable measure across the repeat-length systems. These systems contain different numbers of molecules so molecule-level recurrence would also reflect differences in the number of possible molecular partners. Particle contact recurrence instead measures how often the same individual base pairs reappear across the sampled frames (Figure 14).

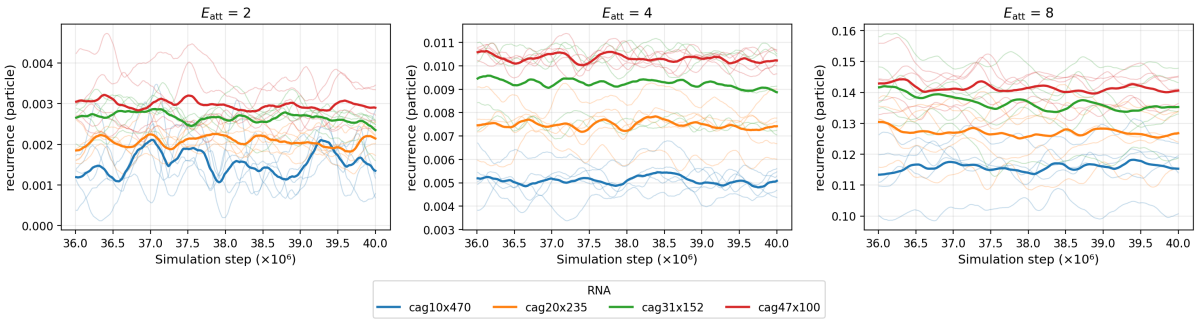


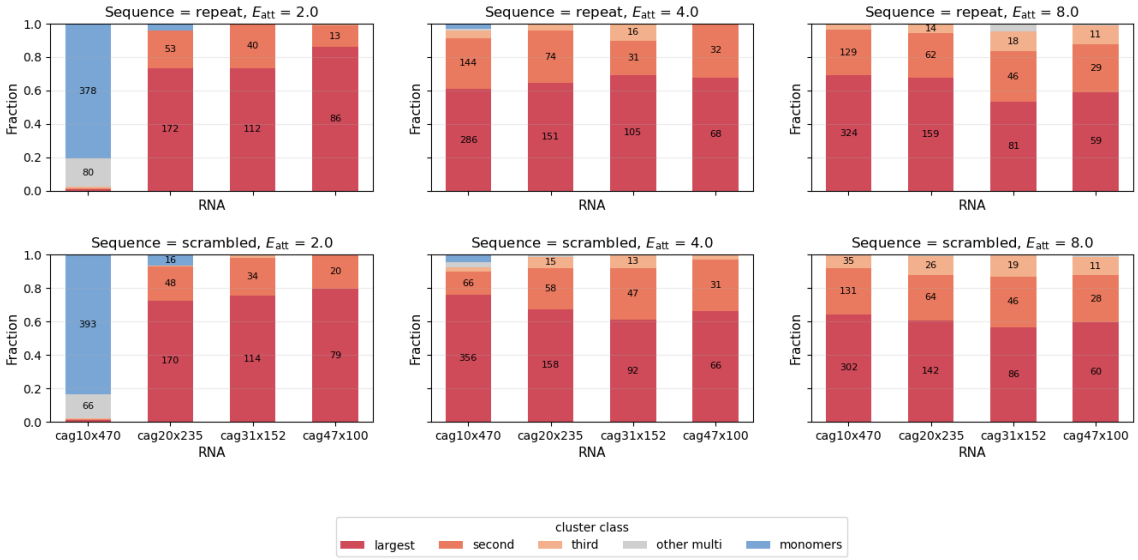
Figure 14: Particle-contact recurrence as a function of simulation step for the four CAG systems at $E_{\mathrm{att}} \in \{2, 4, 8\}$, $\rho = 0.01$, $k_{\mathrm{bbb}} = 2.5$, computed over the final 10% of steps across 5 replicates. Thin lines show individual replicates, thick lines the replicate mean. A value near zero indicates that contacts in the analysis window are formed between continuously changing bead pairs; a value approaching 1 indicates that the same individual bead pairs reappear in every sampled frame.

Recurrence follows the E_{att} trend established in the reentrance sweep: an increase in attraction strength leads to less partner switching and a higher recurrence score. Chain length adds a smaller, consistently ordered effect, with $(\mathrm{CAG})_{47}$ having the highest and $(\mathrm{CAG})_{10}$ the lowest particle recurrence at every attraction strength. Longer chains can maintain several contacts at different positions along the same molecule pair. If one individual base pair breaks while other contacts remain, the two molecules remain spatially associated, increasing the probability that the same bead pair reforms. Shorter chains have fewer such anchoring pairs, so a broken bond is more easily replaced by a new partner, lowering recurrence.

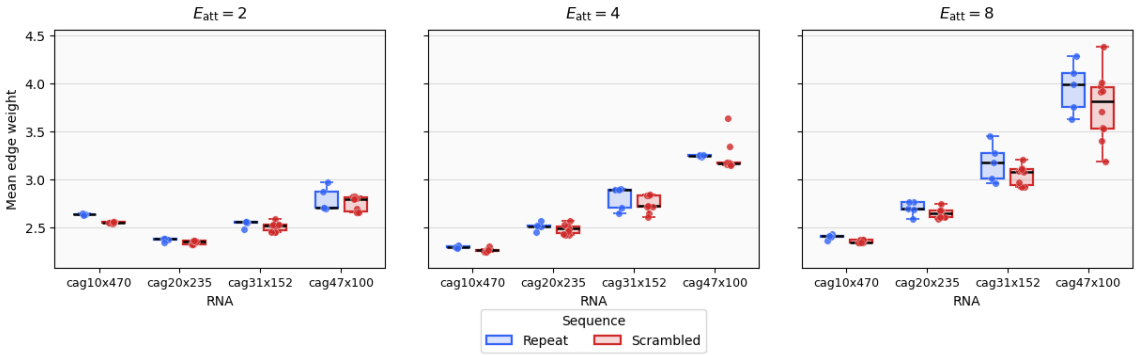
The absolute values remain low throughout, meaning contacts are continually realized by *new* base pairs rather than the same ones reappearing. This low recurrence is expected at the particle level: individual base pairs turn over quickly, so resolving contacts at the particle level exposes this turnover directly.

Clustering propensity is primarily set by length and attraction strength rather than sequence periodicity

To isolate the effect of the periodic CAG pattern, each repeat-length system was compared with a scrambled control matched in molecule number, chain length, and nucleotide composition. The exact procedure for generating the scrambled controls is described in Section 5.3 (Figure 4). Ranked-cluster composition and mean molecule-pair edge weight were compared across both sequence classes (Figure 15). Differences in mean edge weight were assessed from late-time means using two-sided permutation tests, with Holm correction (Figure 15c).



(a) Ranked-cluster composition.



(b) Mean molecule-pair edge weight.

	$E_{att} = 2$	$E_{att} = 4$	$E_{att} = 8$
CAG10	**0.0080****	**0.0366***	**0.0366***
CAG20	0.1259	1.0000	0.9217
CAG31	1.0000	1.0000	1.0000
CAG47	1.0000	1.0000	1.0000

(c) Holm-adjusted permutation-test p-values.

Figure 15: Comparison of periodic CAG repeats with length- and composition-matched scrambled controls. **(a)** Ranked-cluster composition. **(b)** Mean molecule-pair edge weight for qualifying molecule-pair edges. **(c)** Holm-adjusted p-values from two-sided permutation tests comparing replicate-level mean edge weights between repeat ($n = 5$) and scrambled ($n = 10$) systems. Asterisks indicate $p < 0.05$ (*) and $p < 0.01$ (**).

Across conditions, repeat and scrambled sequences showed broadly similar cluster-size distributions (Figure 15a). At $E_{att} = 2$, the strongest difference was associated with chain length rather than sequence patterning: $(CAG)_{10}$ remained predominantly monomeric in both sequence classes, whereas $(CAG)_{20}$, $(CAG)_{31}$, and $(CAG)_{47}$ formed substantially larger assem-

blies. At $E_{\text{att}} = 4$ and 8, multimolecular clusters were present across all tested lengths, with only modest differences in ranked-cluster composition between repeat and scrambled systems.

Mean molecule-pair edge weight showed a stronger dependence on chain length and attraction strength (Figure 15b). Edge weights generally increased with RNA length and, particularly for the longer chains, with increasing E_{att}. Repeat sequences had slightly higher mean edge weights than their scrambled controls in all tested conditions, but these differences were small relative to the overall variation with length and attraction strength. After correction for multiple comparisons, significant repeat-scrambled differences were detected only for $(\text{CAG})_{10}$, at $E_{\text{att}} = 2$, 4, and 8 (Figure 15c). No significant differences were detected for $(\text{CAG})_{20}$, $(\text{CAG})_{31}$, or $(\text{CAG})_{47}$.

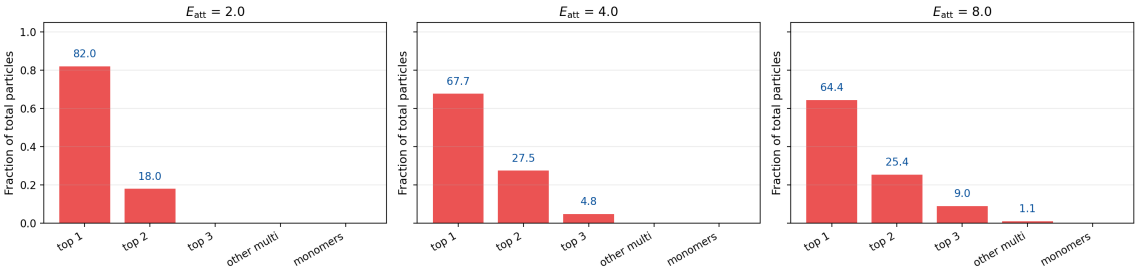
Together, these results indicate that disrupting the periodic CAG arrangement has a limited effect on the overall assembly behavior under the conditions tested. The dominant changes in cluster organization and intermolecular connectivity are instead associated with RNA length and attraction strength, while sequence periodicity produces only a small detectable effect on pairwise connectivity for the shortest chains.

6.4 RQ3: Mixed CAG10/CAG47 systems

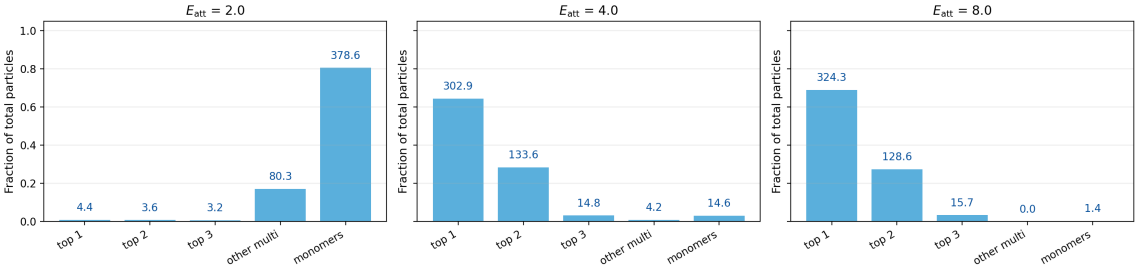
Having established that repeat length affects clustering in single-length systems, the longest and shortest sequences, $(\text{CAG})_{47}$ and $(\text{CAG})_{10}$, were mixed to examine their interaction. The mixture contained 235 molecules of $(\text{CAG})_{10}$ and 50 of $(\text{CAG})_{47}$, thereby matching the nucleotide contribution of each component. As in RQ2, chain stiffness was fixed at the calibrated value of $k_{\text{bbb}} = 2.5$, density at $\rho = 0.01$, and three attraction strengths were tested, $E_{\text{att}} \in \{2, 4, 8\}$.

Long chains promote recruitment of short chains into shared clusters

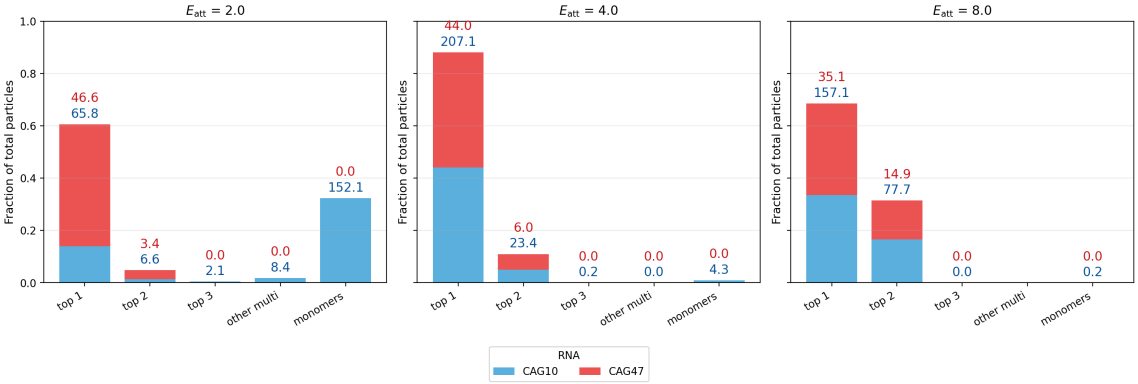
To determine how a 1:1 material mixture differs from pure systems, the mixed system was compared against the pure $(\text{CAG})_{47}$ and pure $(\text{CAG})_{10}$ reference systems from RQ2 at matched density and stiffness (Figure 16).



(a) Pure $(CAG)_{47}$ ranked-cluster composition.



(b) Pure $(CAG)_{10}$ ranked-cluster composition.



(c) Mixed-system cluster material distribution.

Figure 16: Ranked-cluster composition of the pure and mixed systems at $\rho = 0.01$ and $k_{bbb} = 2.5$ across $E_{att} = 2, 4$, and 8. Panels (a) and (b) show the pure $(CAG)_{47}$ and pure $(CAG)_{10}$ reference systems, respectively. Panel (c) shows the mixed $(CAG)_{10}$ +$(CAG)_{47}$ system resolved by RNA type. Bars are grouped by ranked cluster class. In panel (c), colors indicate RNA type, and annotations give the mean number of molecules of each RNA type in the corresponding cluster class. All values are averaged over the final 5% of simulation steps.

The pure reference systems provide the baseline for interpreting the mixed composition (Figure 16c). In the pure $(CAG)_{47}$ system, most material was already clustered at all three attraction strengths, but the distribution across ranked clusters shifted with E_{att} (Figure 16a). At E_{att} = 2, the system was mainly partitioned into two large clusters of 82% and 18% of the total material. At E_{att} = 4 and E_{att} = 8, the largest-cluster fraction decreased to 68% and 64%, respectively, while the second- and third-ranked clusters became more prominent. Thus, even in the pure $(CAG)_{47}$ system, high attraction maintained a clustered state while redistributing material across several large components. By contrast, the pure $(CAG)_{10}$ system was mostly

monomeric at E_{att} = 2 and showed increased clustering into a single large cluster as E_{att} increased (Figure 16b).

Against this reference, the mixed system shows that the presence of $(CAG)_{10}$ does not suppress $(CAG)_{47}$ clustering. Instead, clusters formed predominantly by the longer chains recruit a fraction of the shorter chains (Figure 16c).

At E_{att} = 2, nearly all $(CAG)_{47}$ molecules were found in the two largest clusters, with most of them already concentrated in the largest cluster. In contrast, most $(CAG)_{10}$ molecules remained monomeric. This indicates that, at lower attraction, the largest mixed cluster is primarily a $(CAG)_{47}$-driven network into which only a small fraction of $(CAG)_{10}$ is recruited.

Relative to the pure reference systems at the same total nucleotide density, the mixture was most strongly dominated by a single connected component at E_{att} = 4. The largest cluster contained approximately 88% of total material and included nearly all $(CAG)_{47}$ molecules together with a large fraction of the $(CAG)_{10}$ pool. Compared with the pure $(CAG)_{47}$ system, where material remained distributed between the first- and second-ranked clusters, the mixed system showed stronger recruitment of molecules into a single dominant cluster. This suggests that $(CAG)_{10}$ does not merely dilute the $(CAG)_{47}$ assembly at this attraction strength, but is recruited into and may help connect the dominant cluster.

At E_{att} = 8, the material was redistributed between two large mixed components rather than one dominant cluster. Nearly all material was contained in the two largest clusters, with no monomeric material remaining. This resembles the redistribution seen in the pure $(CAG)_{47}$ system at higher attraction strength, but in the mixture both large clusters contain substantial $(CAG)_{10}$ and $(CAG)_{47}$ contributions. Thus, the apparent reduction in largest-cluster size at E_{att} = 8 reflects reduced merging into a single component, not loss of clustering.

Both chain lengths form a shared contact network

Species-resolved contact analysis shows that $(CAG)_{10}$ and $(CAG)_{47}$ are connected through numerous heterotypic contacts, rather than forming largely separate species-specific networks (Figure 17).

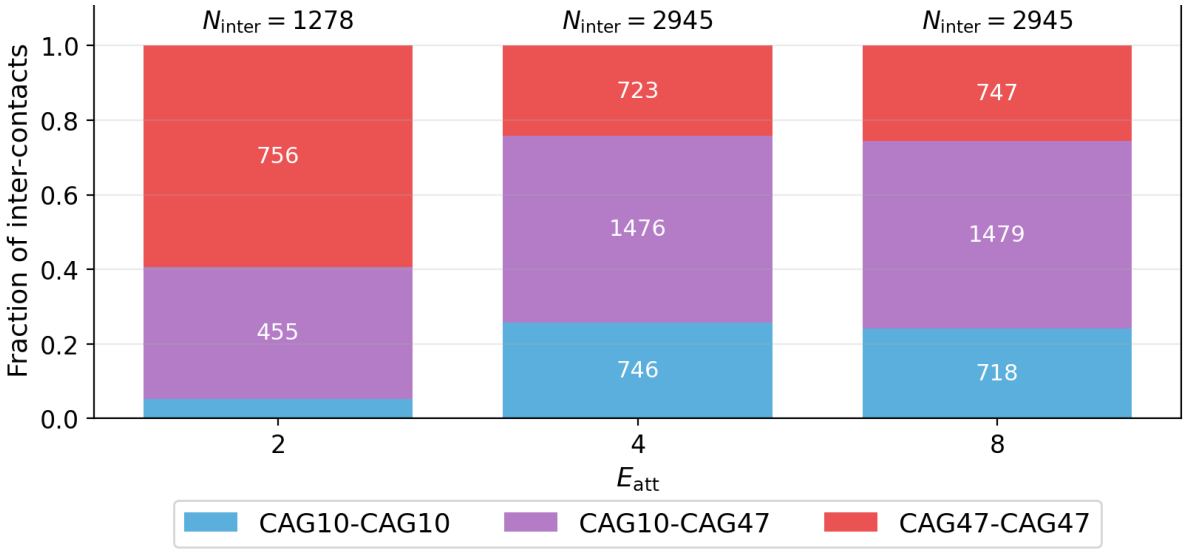


Figure 17: Intermolecular contact composition of the largest cluster over the final 5% of simulation steps. For each E_{att}, stacked bars show the fraction of largest-cluster contacts formed by $(CAG)_{10}$-$(CAG)_{10}$, $(CAG)_{10}$-$(CAG)_{47}$, and $(CAG)_{47}$-$(CAG)_{47}$ molecule pairs. Numbers inside the bars give the mean contact count for each contact type, while N above each bar gives the mean total number of intermolecular contacts in the largest cluster.

At $E_{att} = 2$, the cluster is built around a $(CAG)_{47}$-$(CAG)_{47}$ network ($\sim$756 contacts) decorated by a strong cross-species term ($\sim$455) and almost no $(CAG)_{10}$-$(CAG)_{10}$ bonding. Together with the strong enrichment of long chains in the largest cluster, this is consistent with a long-RNA-enriched network that incorporates a smaller number of short chains.

At $E_{att} = 4$ and $E_{att} = 8$, cross-species $(CAG)_{10}$-$(CAG)_{47}$ contacts formed the largest single contact category, while both homotypic contact types also remained abundant. This contact composition is consistent with extensive mixing of the two RNA species within the largest cluster rather than segregation into separate species-specific networks.

The largest cluster peaks in size but not in connectivity at intermediate attraction

Three properties of the largest cluster were tracked across attraction strengths: the fraction of total particles it contains, the total number of edges in its molecular contact graph and its mean graph degree, i.e. the average number of distinct qualifying molecular partners per molecule (Figure 18).

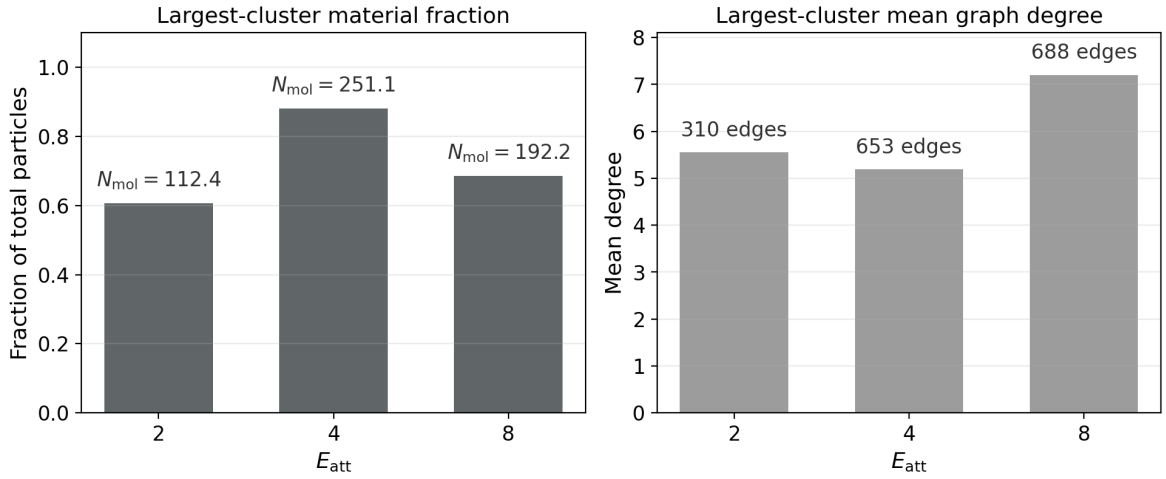


Figure 18: Largest-cluster size and connectivity over the final 5% of simulation steps. The left panel shows the fraction of total system particles contained in the largest cluster, with annotations indicating the mean number of molecules in that cluster. The right panel shows the mean molecule graph degree in the largest cluster and bars are annotated with the mean number of edges.

The size and internal connectivity of the largest cluster responded non-monotonically to attraction strength. The fraction of total system material contained in the largest cluster increased from approximately 0.61 at $E_{att} = 2$ to 0.88 at $E_{att} = 4$, before decreasing to approximately 0.69 at $E_{att} = 8$. These clusters contained, on average, 112.4, 251.1, and 192.2 molecules, respectively. Thus, the most inclusive assembly occurred at $E_{att} = 4$.

The mean graph degree decreased slightly from 5.5 at $E_{att} = 2$ to 5.2 at $E_{att} = 4$, despite the number of qualifying molecule-pair edges increasing from approximately 310 to 653. The number of molecules in the largest cluster therefore increased slightly faster than the number of qualifying edges, producing a larger but marginally less connected network on a per-molecule basis. At $E_{att} = 8$, the largest cluster contained fewer molecules than at $E_{att} = 4$ but had more edges (~ 688) and a substantially higher mean degree (~ 7.2). The most inclusive cluster was therefore not the most highly connected: intermediate attraction maximized recruitment into one component, whereas stronger attraction produced a smaller but more densely connected molecule-level network.

7 Discussion

This thesis used a minimal RNA model to isolate how base-pairing strength, chain stiffness, repeat length, sequence patterning, and mixture composition shape protein-free RNA assembly.

The model recovered several qualitative features expected for RNA. A specific stem-loop structure formed only when base-pair attraction was combined with sufficient stiffness. Multi-chain simulations showed a non-monotonic dependence of clustering on attraction strength, consistent with the reentrant response predicted for saturable associative polymers [41]. Increasing chain length strongly promoted clustering at fixed nucleotide concentration, whereas the periodic CAG arrangement did not produce a consistent advantage over composition-matched scrambled sequences. In mixed-length systems, long RNA chains formed an intermolecular scaffold that recruited shorter chains into a shared network. These findings support the use of the model as a minimal framework for studying the competition between RNA folding and intermolecular association. However, because connected clusters were identified without directly measuring coexistence between dense and dilute phases, the results are best interpreted in terms of clustering and network formation rather than definitive phase separation.

Saturable base pairing produces reentrant RNA clustering

The most prominent collective behavior observed in the simulations was the non-monotonic response of clustering to increasing Watson-Crick attraction strength. At weak attraction, the system remained monomeric because the energetic gain from forming a small number of base pairs was insufficient to compensate for the translational and orientational entropy lost when separate RNA molecules associated. At intermediate attraction strengths, large clusters became strongly populated. At still higher attraction strengths, these large assemblies were suppressed and the system returned to monomers, dimers, trimers and small oligomers. This sequence of weakly associated monomers, large multimers, and strongly bonded small structures is consistent with the three regimes predicted by the analytical enumeration model of Kimchi et al. [41].

The intermediate clustering regime can be understood through the configurational multiplicity of partially bonded assemblies. A large multimer does not correspond to one unique structure. A given number of base pairs may be distributed among many possible binding sites, molecular partners, and combinations of intra- and intermolecular contacts. Large assemblies can therefore realize many distinct connected bonding arrangements while leaving some binding sites unpaired. At intermediate attraction strength, the energetic cost of these unpaired sites remains moderate, allowing the configurational entropy associated with this large number of arrangements to stabilize multimeric states. This extends the conventional sticker-spacer picture, in which multivalency promotes association, by emphasizing that the statistical weight of

an assembly depends not only on how many bonds it contains but also on how many ways those bonds can be arranged [41].

At high attraction strength, the energetic penalty of leaving a complementary site unpaired becomes increasingly important. The system is consequently driven toward states in which a large fraction of the finite bonding capacity is satisfied. Small folded monomers, dimers, and oligomers can achieve extensive bond saturation without recruiting many additional molecules and therefore avoid much of the translational and orientational cost of forming a large assembly. Full or near-full saturation also restricts the number of accessible bonding arrangements, reducing the configurational-entropy advantage of partially bonded large multimers. The large-cluster state is therefore destabilized even though individual base-pairing interactions have become stronger.

The contact analysis directly supports this interpretation. The total number of base-pair contacts approached a plateau at higher E_{att}, showing that the reentrant state was not produced by a loss of bonding. Instead, contacts were redistributed from predominantly intermolecular at intermediate attraction to predominantly intramolecular at high attraction. The simultaneous reduction in intermolecular edge count and retention of high molecule-pair edge weights further shows that the remaining intermolecular associations were not weak. Rather, fewer molecule pairs remained connected, but those that did were reinforced by many simultaneous base pairs.

The high-attraction regime is more accurately described as a loss of large-scale intermolecular connectivity than as complete dissolution. Strong attraction promotes local bond saturation and persistent small structures while suppressing the broad distribution of bonds across many molecular partners that is required for a large-scale connected network. This distinction is important because cluster size alone could otherwise suggest that the system simply becomes weakly associated again at high E_{att}. In fact, the microscopic origins of the weak- and strong-attraction monomeric regimes are different: weak attraction produces largely unbound and dynamic molecules, whereas strong attraction produces internally saturated or strongly oligomerized molecules with reduced capacity to connect to additional partners.

Chain stiffness redirects bonding between folding and assembly

The single-chain and multi-chain simulations together show that stiffness controls how complementary sites are distributed between specific folding, nonspecific collapse, and intermolecular association. In the single-chain system, increasing attraction strength alone did not guarantee recovery of the ViennaRNA reference structure. Fully flexible chains formed increasingly compact conformations as E_{att} increased, but these structures contained numerous non-reference contacts. Their low radius of gyration reflected nonspecific collapse rather than correct folding. Some angular stiffness was required to restrict the conformational search space and suppress geometrically accessible but incorrectly registered contacts. These results demonstrate that

attraction strength primarily determines whether base pairs form at all, whereas stiffness contributes to the specificity of those contacts.

The same geometric mechanism affects intermolecular clustering. Flexible chains can fold back on themselves at little bending cost, allowing complementary sites on the same molecule to satisfy one another. These intramolecular loops consume the finite bonding capacity without contributing to connectivity between molecules, as anticipated by associative polymer theories that incorporate loop formation [42]. Increasing stiffness suppresses the geometrically accessible self-folding and therefore reallocates a greater fraction of bonds to intermolecular partners. This effect was especially strong at weak and intermediate attraction, where increasing k_{bbb} substantially raised the intermolecular contact fraction.

Importantly, stiffness also changed the internal architecture of clusters. At intermediate attraction strengths, fully flexible and calibrated-stiffness chains showed similar large-cluster populations, but the latter formed substantially more molecule-pair edges. Clusters of comparable size were therefore more sparsely connected when the chains were flexible. By shifting bonding from intramolecular loops toward intermolecular contacts, moderate stiffness increased the number of distinct molecular connections without necessarily changing cluster size.

The response was nevertheless not monotonic across all stiffness values. The calibrated stiffness produced the broadest large-cluster window, while the stiffest chains at $k_{\mathrm{bbb}} = 5$ exited this regime at lower attraction strengths and formed a greater proportion of dimers, trimers, and small oligomers. One possible explanation is that very stiff chains cannot easily form compact self-saturated monomers but can align with one or a few intermolecular partners and form several cooperative base pairs. At the same time, excessive stiffness may limit the chain geometries that permit simultaneous bonding to several differently positioned partners. This could favor strongly bonded small oligomers over broadly connected multimeric networks.

This provides an extension to the reentrant theory for saturable polymers [41]. The position and architecture of the reentrant regime depend not only on sticker number and interaction strength but also on the conformational cost of arranging those stickers in intra- and intermolecular geometries. In physical RNA, persistence length varies with sequence, base stacking, and ionic conditions [45]. Changes in ionic environment may therefore affect condensation by altering electrostatic screening or pairing energy but also by modifying chain geometry and the relative accessibility of competing bonding arrangements.

RNA length lowers the interaction threshold for network formation

Increasing CAG-repeat length strongly increased clustering propensity at fixed total particle count and density, effectively lowering the attraction strength required to cross the threshold into network formation. The shortest system, $(\mathrm{CAG})_{10}$, remained predominantly monomeric at $E_{\mathrm{att}} = 2$, whereas longer chains formed large clusters. At $(\mathrm{CAG})_{20}$, the monomer fraction

already decreased substantially and large connected assemblies emerged. By $(\mathrm{CAG})_{31}$ and $(\mathrm{CAG})_{47}$, most material was incorporated into large clusters under the same conditions. This qualitative length dependence is consistent with experimental observations that longer RNAs are preferentially enriched in stress granules and that expanded CAG and CUG repeat RNAs show a threshold-like increase in condensation at approximately 30 repeats [20, 17, 47]. It also agrees with previous coarse-grained simulations in which longer repeat RNAs formed more extensive intermolecular networks [26, 31, 30].

Two related polymer mechanisms can account for this result. First, the translational entropy contribution of a polymer to the free energy of mixing decreases with increasing degree of polymerization. In the Flory-Huggins description, the entropic term scales as $\phi/N \ln \phi$, so dispersing a fixed number of nucleotides as many short chains provides a larger mixing entropy than dispersing the same material as fewer long chains [37, 38]. Association of long chains therefore incurs a smaller translational entropy penalty per nucleotide.

Second, longer chains have higher molecular valency. They contain more complementary sites and can form several spatially separated intermolecular contacts, allowing one RNA molecule to bridge multiple partners. A short chain may form a strongly bonded pair but still have insufficient remaining valency to extend the interaction into a larger network. A long chain can retain contacts with one molecule while simultaneously recruiting others, lowering the connectivity threshold for the formation of a large component. Longer chains also form higher-weight molecular edges because they can support more simultaneous base pairs between the same pair of molecules.

The length required for substantial clustering shifted with attraction strength. At $E_{\mathrm{att}} = 2$, the distinction between $(\mathrm{CAG})_{10}$ and longer chains was strongest. At $E_{\mathrm{att}} = 4$, clustering extended to shorter sequences, and at $E_{\mathrm{att}} = 8$ even $(\mathrm{CAG})_{10}$ formed substantial assemblies. Shorter chains have lower molecular valency and consequently can form fewer intermolecular contacts per molecule and depend more heavily on per-contact energy to compensate. Raising E_{att} increases the energetic gain from each contact enough to permit network formation even at low degrees of polymerization. Chain length and base-pairing strength thus act jointly in setting the onset of large cluster formation.

Sequence periodicity has a limited effect on clustering in the minimal CG model

In contrast to the pronounced effect of chain length, the periodic CAG arrangement produced no consistent clustering advantage over composition-matched scrambled controls. Repeat and scrambled sequences showed broadly similar cluster-size distributions across the tested conditions. Mean molecule-pair edge weights were slightly higher for the repeat sequences, with statistically significant repeat–scrambled differences detected only for $(\mathrm{CAG})_{10}$ while no sig-

nificant differences were detected for $(CAG)_{20}$, $(CAG)_{31}$ or $(CAG)_{47}$. The regular CAG pattern was therefore not required for large-cluster formation within the present model, although a small effect on pairwise intermolecular connectivity remained detectable for the shortest chains.

This result differs from studies in which CAG-repeat RNA showed stronger condensation than composition-matched scrambled sequences [20, 26]. However, other experimental work has reported that scrambled CAG sequences can also phase-separate and that the direction of the periodicity effect may depend on repeat length and experimental conditions [52]. The present result should therefore not be interpreted as evidence that sequence patterning is generally irrelevant. Instead, it identifies which aspects of sequence patterning are not captured by the tested parameters and level of coarse-graining.

The weak effect of periodicity observed in this thesis may be specific to the tested chain stiffness of $k_{bbb} = 2.5$. More flexible chains can rearrange at relatively little conformational cost to accommodate irregularly spaced complementary sites, potentially reducing the importance of their periodic patterning. As stiffness increases, such rearrangements become more constrained and regular sticker ordering may have a more pronounced stabilizing effect on clustering. Periodic CAG chains may exhibit a higher configurational entropy than scrambled chains when forming a cluster, as the regular sticker architecture permits more alternative bonding arrangements. Testing the scrambled controls across additional stiffness values would be required to determine whether such an effect emerges.

RNA features omitted from the model may also reduce the effects of repeat periodicity. In the model, each nucleotide is represented by a single isotropic bead, and all complementary G-C pairs are assigned the same interaction strength regardless of neighboring bases or relative orientation. Because CAG contains no uracil, the adenine beads do not participate in base pairing in these systems. The effective attractive sticker content is therefore determined primarily by the total number of G and C beads, which is preserved during scrambling. Sequence order changes the spacing of these stickers along the chain but does not change the total bonding capacity or introduce cooperative energetic effects between adjacent base pairs.

In RNA, consecutive G-C pairs can be stabilized by base stacking and nearest-neighbor interactions, while helical geometry imposes directional constraints on the alignment of complementary regions. These structures depend on local orientation, stacking cooperativity, loop geometry, and sequence-context-dependent energetics. An isotropic pair potential can reproduce complementary contact formation but cannot fully distinguish a continuous, cooperatively aligned duplex segment from the same number of independently distributed G-C contacts.

The absence of a periodicity effect is thus informative about the limits of a sticker-count description. Complementary sticker number, saturability, stiffness, and chain length are sufficient to generate strong length-dependent and reentrant clustering, but they are not sufficient to generate a robust advantage for the periodic CAG register. Recovering this effect may require

directional base-pairing potentials, explicit stacking or sequence-dependent nearest-neighbor energies. Future model extensions could test these mechanisms individually to determine which physical feature primarily contributes to the experimentally observed sensitivity to repeat patterning.

Long RNA recruits short RNA into shared molecular networks

The mixed $(CAG)_{10}$/$(CAG)_{47}$ simulations demonstrate that short RNA chains that cluster only weakly in isolation can nevertheless become incorporated into assemblies formed by longer RNA. The extent and architecture of this recruitment depended on attraction strength.

At $E_{att} = 2$, nearly all $(CAG)_{47}$ molecules were incorporated into the two largest clusters, whereas most $(CAG)_{10}$ molecules remained monomeric. Within the largest cluster, $(CAG)_{47}$-$(CAG)_{47}$ contacts formed the largest contact contribution, while a substantial but smaller number of $(CAG)_{10}$-$(CAG)_{47}$ contacts connected short chains to the $(CAG)_{47}$ enriched assembly. Very little $(CAG)_{10}$-$(CAG)_{10}$ intermolecular bonding occurred. This indicates that the long chains formed the dominant structural component of the connected assembly, while for $(CAG)_{10}$ molecules, the translational-entropy gain of remaining unbound still outweighed the free-energy gain from joining the cluster.

At $E_{att} = 4$, recruitment of the short chains increased substantially. The largest cluster contained approximately 88% of the total system material, including nearly all long chains and a large fraction of the $(CAG)_{10}$ population. Cross-species contacts became the largest single contact category, exceeding both homotypic contributions. The mixture therefore did not consist of separate $(CAG)_{10}$- and $(CAG)_{47}$-rich clusters joined by only a few bridges. Instead, both chain lengths participated throughout a shared intermolecular contact network. This does not by itself imply preferential heterotypic binding, because the expected abundance of cross-species contacts also increases as the two species become more evenly represented within the cluster. Rather, the contact composition shows that the two RNA lengths are extensively intermixed within a shared molecular network rather than segregating into separate species-specific assemblies.

At $E_{att} = 8$, essentially no monomeric material remained, but the system was divided between two large mixed clusters rather than forming one dominant component. The decrease in the largest-cluster fraction relative to $E_{att} = 4$ produced a redistribution of material among multiple highly connected components. This is consistent with the broader observation that increasing attraction does not monotonically improve coalescence. Stronger bonding may reduce cluster merging or restructuring once strongly bonded assemblies have formed, particularly when molecular partners become persistent.

A plausible explanation for the recruitment of $(CAG)_{10}$ is the difference in molecular valency between the two chain lengths. The higher-valency $(CAG)_{47}$ chains can bind several molecular

partners simultaneously and therefore sustain a connected intermolecular network where, under the same conditions, $(\mathrm{CAG})_{10}$ cannot do so independently. Once this long-chain network is established, a short chain does not need to provide enough valency to propagate connectivity itself; it only needs to form sufficiently favorable contacts with an already connected component. The loss of translational entropy upon recruitment can then be offset by the binding free energy of these contacts and, at intermediate attraction strength, by the high configurational multiplicity of the large multimer, which permits many alternative bonding arrangements for incorporating additional chains.

The mixed system also illustrates that cluster size and internal connectivity are distinct properties. The largest cluster reached its greatest material fraction at $E_{\mathrm{att}} = 4$, but its mean molecular degree was higher at $E_{\mathrm{att}} = 8$. Intermediate attraction therefore maximized recruitment into one inclusive assembly, whereas stronger attraction produced a smaller but more densely connected network. A large cluster can incorporate many molecules through a relatively sparse set of links, while a smaller cluster can contain more interaction partners per molecule.

These findings complement previous work on mixed-length RNA-protein condensates, in which long RNAs preferentially occupy network-forming regions while short RNAs can accumulate at interfaces [53]. The present simulations do not resolve radial partitioning or interfacial localization and therefore cannot establish a surfactant role for $(\mathrm{CAG})_{10}$. They nevertheless support a more general principle: assembly of a low-valency RNA can be promoted by the presence of a higher-valency RNA that supplies the connected scaffold. In cellular condensates, RNA enrichment may depend not only on the intrinsic phase-separation propensity of an individual transcript but also on its ability to form contacts with other, more strongly network-forming RNAs.

Model limitations and future directions

The simplicity of the model enables systematic variation of individual parameters, but also limits the biological conclusions that can be drawn. First, clustering was quantified through molecular contact graphs and connected components. These analyses establish molecular association and network formation but do not demonstrate equilibrium coexistence between dense and dilute liquid phases. Direct characterization of LLPS would require concentration-dependent simulations, measurement of dilute- and dense-phase concentrations, and ideally construction of coexistence curves.

Second, the single-bead representation omits several physical features of RNA. The sugar-phosphate backbone, explicit electrostatics, counterions, base stacking, noncanonical interactions, and directional hydrogen-bond geometry are not represented. The attraction parameter E_{att} is therefore a generic reduced interaction strength and cannot be mapped directly onto experimental base-pairing free energy, temperature, or ion concentration. Future work

could extend the model to a two-bead representation separating the nucleobase from the sugar-phosphate backbone, allowing electrostatic effects to be introduced through implicit screened backbone-backbone interactions.

Third, A-U and G-C pairs were assigned the same interaction strength, despite their different physical stabilities. Base-pairing valency was imposed through a density-dependent potential rather than directional bonding. Although this successfully limited excessive simultaneous contacts, it represents valency approximately and may permit local geometries that would be inaccessible in physical RNA. These simplifications are particularly relevant to the sequence periodicity analysis, where stacking and nearest-neighbor cooperativity may be required to distinguish repeat and scrambled sequences. A two-bead representation, as suggested above, could provide additional orientational information through bonded angle or dihedral terms and support directional base-pairing and stacking interactions. Local sequence context could also be incorporated into the interaction strength, as in RNA2PS [30], allowing neighboring bases to influence the stability of a contact rather than treating each complementary pair independently.

Fourth, the single-chain validation used one designed hairpin and computational reference structures. Agreement with the ViennaRNA MFE pair list and the AlphaFold3 radius of gyration demonstrates that the model can recover one target conformation under selected parameters, but does not constitute general validation across diverse RNA folds. Future work should test multiple hairpins, duplexes, bulges, and repeat-containing structures, ideally against experimentally determined ensembles.

Finally, the mixed-length analysis demonstrated shared networks but did not quantify spatial partitioning within the clusters. Radial density profiles and surface enrichment would be needed to distinguish preferential recruitment, random incorporation, core localization, or interfacial behavior. Extending the mixture simulations across additional stiffness values, chain lengths and compositions would also help determine whether short RNA generally enhances coalescence or whether the observed response is restricted to particular length ratios and interaction strengths.

Conclusion

Despite its limitations, the minimal model reveals a coherent physical picture. RNA assembly is governed not simply by the total number or strength of base pairs, but by how a finite bonding capacity is distributed among competing conformations and molecular partners. Weak interactions leave chains dispersed, intermediate interactions allow many partially bonded network arrangements, and strong interactions favor locally saturated structures. Persistence length selects which of these arrangements are geometrically accessible, while chain length determines whether a molecule can extend association into a larger network. In mixed systems, long chains can provide this network-forming capacity and recruit shorter chains that remain largely dispersed on their own. These results demonstrate how relatively simple polymer properties can combine to produce complex and non-monotonic RNA assembly behavior.

References

[1] S. F. Banani, H. O. Lee, A. A. Hyman, and M. K. Rosen. Biomolecular condensates: Organizers of cellular biochemistry. In: *Nature Reviews Molecular Cell Biology* 18.5 (2017), pp. 285–298. DOI: 10.1038/nrm.2017.7.

[2] F. Frottin, F. Schueder, S. Tiwary, R. Gupta, R. Körner, T. Schlichthaerle, J. Cox, R. Jungmann, F. U. Hartl, and M. S. Hipp. The nucleolus functions as a phase-separated protein quality control compartment. In: *Science* 365.6451 (2019), pp. 342–347. DOI: 10.1126/science.aaw9157.

[3] D. Campos-Melo, Z. C. E. Hawley, C. A. Droppelmann, and M. J. Strong. The Integral Role of RNA in Stress Granule Formation and Function. In: *Frontiers in Cell and Developmental Biology* 9:621779 (2021). DOI: 10.3389/fcell.2021.621779.

[4] Y. Luo, Z. Na, and S. A. Slavoff. P-Bodies: Composition, Properties, and Functions. In: *Biochemistry* 57.17 (2018), pp. 2424–2431. DOI: 10.1021/acs.biochem.7b01162.

[5] A. Zbinden, M. Pérez-Berlanga, P. De Rossi, and M. Polymenidou. Phase Separation and Neurodegenerative Diseases: A Disturbance in the Force. In: *Developmental Cell* 55.1 (2020), pp. 45–68. DOI: 10.1016/j.devcel.2020.09.014.

[6] P. Li, S. Banjade, H.-C. Cheng, S. Kim, B. Chen, L. Guo, M. Llaguno, J. V. Hollingsworth, D. S. King, and S. F. Banani. Phase transitions in the assembly of multivalent signalling proteins. In: *Nature* 483 (2012), pp. 336–340. DOI: 10.1038/nature10879.

[7] G. L. Dignon, R. B. Best, and J. Mittal. Biomolecular Phase Separation: From Molecular Driving Forces to Macroscopic Properties. In: *Annual Review of Physical Chemistry* 71.1 (2020), pp. 53–75. DOI: 10.1146/annurev-physchem-071819-113553.

[8] C. P. Brangwynne, P. Tompa, and R. V. Pappu. Polymer physics of intracellular phase transitions. In: *Nature Physics* 11.11 (2015), pp. 899–904. DOI: 10.1038/nphys3532.

[9] Y.-H. Lin, J. D. Forman-Kay, and H. S. Chan. Theories for Sequence-Dependent Phase Behaviors of Biomolecular Condensates. In: *Biochemistry* 57.17 (2018), pp. 2499–2508. DOI: 10.1021/acs.biochem.8b00058.

[10] A. Molliex, J. Temirov, J.-H. Lee, M. Coughlin, A. P. Kanagaraj, H.-J. Kim, T. Mittag, and J. P. Taylor. Phase separation by low complexity domains promotes stress granule assembly and drives pathological fibrillization. In: *Cell* 163.1 (2015), pp. 123–133. DOI: 10.1016/j.cell.2015.09.015.

[11] Y. Lin, D. S. W. Protter, M. K. Rosen, and R. Parker. Formation and Maturation of Phase Separated Liquid Droplets by RNA Binding Proteins. In: *Molecular cell* 60.2 (2015), pp. 208–219. DOI: 10.1016/j.molcel.2015.08.018.

[12] A. Patel, H.-H. Lee, L. Jawerth, S. Maharana, M. Jahnel, M. Y. Hein, S. Stoynov, J. Mahamid, S. Saha, and T. M. Franzmann. A liquid-to-solid phase transition of the ALS protein FUS accelerated by disease mutation. In: *Cell* 162.5 (2015), pp. 1066–1077. DOI: 10.1016/j.cell.2015.07.047.

[13] S. Maharana, J. Wang, D. K. Papadopoulos, D. Richter, A. Pozniakovsky, I. Poser, M. Bickle, S. Rizk, J. Guillén-Boixet, and T. M. Franzmann. RNA buffers the phase separation behavior of prion-like RNA binding proteins. In: *Science* 360.6391 (2018), pp. 918–921. DOI: 10.1126/science.aar7366.

[14] T. Ogura, U. Matsuura, M. Machida, K. Nagai, S. Kajimoto, S. Tahara, and T. Nakabayashi. RNA-Mediated Inhibition Mechanism of Liquid–Liquid Phase Separation and Subsequent Aggregation Revealed by Raman Microscopy. In: *JACS Au* 5.11 (2025), pp. 5749–5757. DOI: 10.1021/jacsau.5c01234.

[15] B. Van Treeck, D. S. W. Protter, T. Matheny, A. Khong, C. D. Link, and R. Parker. RNA self-assembly contributes to stress granule formation and defining the stress granule transcriptome. In: *Proceedings of the National Academy of Sciences* 115.11 (2018), pp. 2734–2739. DOI: 10.1073/pnas.1800038115.

[16] F. Pir Cakmak, A. William M. Jr., and C. D. Keating. RNA-Based Coacervates as a Model for Membraneless Organelles: Formation, Properties, and Interfacial Liposome Assembly. In: *Langmuir* 32.39 (2016), pp. 10042–10053. DOI: 10.1021/acs.langmuir.6b02499.

[17] B. Van Treeck and R. Parker. Emerging Roles for Intermolecular RNA-RNA Interactions in RNP Assemblies. In: *Cell* 174.4 (2018), pp. 791–802. DOI: 10.1016/j.cell.2018.07.023.

[18] Y.-B. Lee, H.-J. Chen, J. N. Peres, et al. Hexanucleotide Repeats in ALS/FTD Form Length-Dependent RNA Foci, Sequester RNA Binding Proteins, and Are Neurotoxic. In: *Cell Reports* 5.5 (2013), pp. 1178–1186. DOI: 10.1016/j.celrep.2013.10.049.

[19] J. P. T. Ouyang, S. Shukla, M. Bensalah, and R. Parker. DM1 repeat-expanded RNAs confer RNA toxicity as individual nuclear-retained RNAs. In: *Cell Reports* 44.5:115582 (2025). DOI: 10.1016/j.celrep.2025.115582.

[20] A. Jain and R. D. Vale. RNA phase transitions in repeat expansion disorders. In: *Nature* 546.7657 (2017), pp. 243–247. DOI: 10.1038/nature22386.

[21] Y. Sun, H. Dai, X. Dai, et al. m1A in CAG repeat RNA binds to TDP-43 and induces neurodegeneration. In: *Nature* 623.7987 (2023), pp. 580–587. DOI: 10.1038/s41586-023-06701-5.

[22] K. J. Rohilla and K. T. Gagnon. RNA biology of disease-associated microsatellite repeat expansions. In: *Acta Neuropathologica Communications* 5.1 (2017), p. 63. DOI: 10.1186/s40478-017-0468-y.

[23] C. M. Rodriguez and P. K. Todd. New pathologic mechanisms in nucleotide repeat expansion disorders. In: *Neurobiology of Disease* 130 (2019), p. 104515. DOI: 10.1016/j.nbd.2019.104515.

[24] R. Anderson, M. R. Das, Y. Chang, K. Farenhem, C. O. Schmitz, and A. Jain. CAG repeat expansions create splicing acceptor sites and produce aberrant repeat-containing RNAs. In: *Molecular Cell* 84.4 (2024), 702–714.e10. DOI: 10.1016/j.molcel.2024.01.006.

[25] L. M. Castelli, W.-P. Huang, Y.-H. Lin, K.-Y. Chang, and G. M. Hautbergue. Mechanisms of repeat-associated non-AUG translation in neurological microsatellite expansion disorders. In: *Biochemical Society Transactions* 49.2 (2021), pp. 775–792. DOI: 10.1042/BST20200690.

[26] H. T. Nguyen, N. Hori, and D. Thirumalai. Condensates in RNA repeat sequences are heterogeneously organized and exhibit reptation dynamics. In: *Nature Chemistry* 14 (2022), pp. 775–785. DOI: 10.1038/s41557-022-00934-z.

[27] L. Wan, C. Zhang, Y. Liu, J. Zhang, L. He, Y. Zhang, G. Chen, P. Guo, and D. Han. Small CAG Repeat RNA Forms a Duplex Structure with Sticky Ends That Promote RNA Condensation. In: *Journal of the American Chemical Society* 147.4 (2025), pp. 3813–3822. DOI: 10.1021/jacs.4c16886.

[28] Y. Pan, J. Lu, X. Feng, et al. Gelation of cytoplasmic expanded CAG RNA repeats suppresses global protein synthesis. In: *Nature Chemical Biology* 19.11 (2023), pp. 1372–1383. DOI: 10.1038/s41589-023-01384-5.

[29] I. Yasuda, S. von Bülow, G. Tesei, E. Yamamoto, K. Yasuoka, and K. Lindorff-Larsen. Coarse-grained model of disordered RNA for simulations of biomolecular condensates. In: *Journal of Chemical Theory and Computation* 21.5 (2025), pp. 2766–2779. DOI: 10.1021/acs.jctc.4c01646.

[30] A. R. Tejedor, J. Luengo-Márquez, J. Oller-Iscar, J. Ramirez, A. Ocana, R. Collepardo-Guevara, P. Llombart, and J. R. Espinosa. *RNA2PS: A sequence-specific coarse-grained RNA model linking structure, thermodynamics and phase separation*. 2026. DOI: 10.64898/2026.04.24.720717.

[31] S. Li and J. Chen. Driving forces of RNA condensation revealed through coarse-grained modeling with explicit Mg2+. In: *Proceedings of the National Academy of Sciences* 122.43 (2025), e2504583122. DOI: 10.1073/pnas.2504583122.

[32] D. Yangaliev and S. B. Ozkan. Coarse-grained RNA model for the Martini 3 force field. In: *Biophysical Journal* 125.2 (2026), pp. 445–456. DOI: 10.1016/j.bpj.2025.07.034.

[33] P. Šulc, F. Romano, T. E. Ouldridge, J. P. K. Doye, and A. A. Louis. A nucleotide-level coarse-grained model of RNA. In: *The Journal of Chemical Physics* 140.23 (2014), p. 235102. DOI: 10.1063/1.4881424.

[34] M. Rubinstein and R. H. Colby. *Polymer Physics*. OUP Oxford, 2003.

[35] T. S. Harmon, A. S. Holehouse, M. K. Rosen, and R. V. Pappu. Intrinsically disordered linkers determine the interplay between phase separation and gelation in multivalent proteins. In: *eLife* 6 (2017). Ed. by A. A. Hyman, e30294. DOI: 10.7554/eLife.30294.

[36] K. A. Dill and S. Bromberg. *Molecular driving forces: statistical thermodynamics in biology, chemistry, physics, and nanoscience*. 2nd ed. Boca Raton, Florida: CRC press, Taylor & Francis group, 2011.

[37] P. J. Flory. Thermodynamics of High Polymer Solutions. In: *The Journal of Chemical Physics* 10.1 (1942), pp. 51–61. DOI: 10.1063/1.1723621.

[38] M. L. Huggins. Solutions of Long Chain Compounds. In: *The Journal of Chemical Physics* 9.5 (1941), p. 440. DOI: 10.1063/1.1750930.

[39] J.-M. Choi, A. A. Hyman, and R. V. Pappu. Generalized models for bond percolation transitions of associative polymers. In: *Physical Review E* 102.4 (2020), p. 042403. DOI: 10.1103/PhysRevE.102.042403.

[40] Q. Vicens and J. S. Kieft. Thoughts on how to think (and talk) about RNA structure. In: *Proceedings of the National Academy of Sciences* 119.17 (2022), e2112677119. DOI: 10.1073/pnas.2112677119.

[41] O. Kimchi, E. M. King, and M. P. Brenner. Uncovering the mechanism for aggregation in repeat expanded RNA reveals a reentrant transition. In: *Nature Communications* 14.1 (2023), p. 332. DOI: 10.1038/s41467-023-35803-x.

[42] A. N. Semenov and M. Rubinstein. Thermoreversible Gelation in Solutions of Associative Polymers. 1. Statics. In: *Macromolecules* 31.4 (1998), pp. 1373–1385. DOI: 10.1021/ma970616h.

[43] M. Broda, E. Kierzek, Z. Gdaniec, T. Kulinski, and R. Kierzek. Thermodynamic Stability of RNA Structures Formed by CNG Trinucleotide Repeats. Implication for Prediction of RNA Structure. In: *Biochemistry* 44.32 (2005), pp. 10873–10882. DOI: 10.1021/bi0502339.

[44] H. Maity, H. T. Nguyen, N. Hori, and D. Thirumalai. Odd–even disparity in the population of slipped hairpins in RNA repeat sequences with implications for phase separation.

In: *Proceedings of the National Academy of Sciences* 120.24 (2023), e2301409120. DOI: 10.1073/pnas.2301409120.

[45] H. Chen, S. P. Meisburger, S. A. Pabit, J. L. Sutton, W. W. Webb, and L. Pollack. Ionic strength-dependent persistence lengths of single-stranded RNA and DNA. In: *Proceedings of the National Academy of Sciences* 109.3 (2012), pp. 799–804. DOI: 10.1073/pnas.1119057109.

[46] I. Malhotra and D. A. Potoyan. Re-entrant transitions of locally stiff RNA chains in the presence of polycations leads to gelated architectures. In: *Soft Matter* 19.29 (2023), pp. 5622–5629. DOI: 10.1039/D3SM00320E.

[47] A. Khong, T. Matheny, S. Jain, S. F. Mitchell, J. R. Wheeler, and R. Parker. The stress granule transcriptome reveals principles of mRNA accumulation in stress granules. In: *Molecular cell* 68.4 (2017), 808–820.e5. DOI: 10.1016/j.molcel.2017.10.015.

[48] J. K. A. Tom, P. L. Onuchic, and A. A. Deniz. Short PolyA RNA Homopolymers Undergo Mg2+ -Mediated Kinetically Arrested Condensation. In: *The Journal of Physical Chemistry B* 126.46 (2022), pp. 9715–9725. DOI: 10.1021/acs.jpcb.2c05935.

[49] V. Ramachandran and D. A. Potoyan. Energy landscapes of homopolymeric RNAs revealed by deep unsupervised learning. In: *Biophysical Journal* 123.9 (2024), pp. 1152–1163. DOI: 10.1016/j.bpj.2024.04.003.

[50] D. H. Mathews, J. Sabina, M. Zuker, and D. H. Turner. Expanded sequence dependence of thermodynamic parameters improves prediction of RNA secondary structure1. In: *Journal of Molecular Biology* 288.5 (1999), pp. 911–940. DOI: 10.1006/jmbi.1999.2700.

[51] D. Aierken and J. A. Joseph. Accelerated Simulations Reveal Physicochemical Factors Governing Stability and Composition of RNA Clusters. In: *Journal of Chemical Theory and Computation* 20.22 (2024), pp. 10209–10222. DOI: 10.1021/acs.jctc.4c00803.

[52] G. M. Wadsworth, W. J. Zahurancik, X. Zeng, P. Pullara, L. B. Lai, V. Sidharthan, R. V. Pappu, V. Gopalan, and P. R. Banerjee. RNAs undergo phase transitions with lower critical solution temperatures. In: *Nature Chemistry* 15.12 (2023), pp. 1693–1704. DOI: 10.1038/s41557-023-01353-4.

[53] I. Sanchez-Burgos, L. Herriott, R. Collepardo-Guevara, and J. R. Espinosa. Surfactants or scaffolds? RNAs of varying lengths control the thermodynamic stability of condensates differently. In: *Biophysical Journal* 122.14 (2023), pp. 2973–2987. DOI: 10.1016/j.bpj.2023.03.006.

[54] J. A. Anderson, J. Glaser, and S. C. Glotzer. HOOMD-blue: A Python package for high-performance molecular dynamics and hard particle Monte Carlo simulations. In: *Com-*

putational Materials Science 173 (2020), p. 109363. DOI: 10.1016/j.commatsci.2019.109363.

[55] S. Neidle and M. Sanderson. *Principles of Nucleic Acid Structure*. Academic Press, 2021.

[56] S. Majumder, S. Coupe, N. Fakhri, and A. Jain. Sequence-encoded intermolecular base pairing modulates fluidity in DNA and RNA condensates. In: *Nature Communications* 16.1 (2025), p. 4258. DOI: 10.1038/s41467-025-59456-0.

[57] R. Lorenz, S. H. Bernhart, C. Höner zu Siederdissen, H. Tafer, C. Flamm, P. F. Stadler, and I. L. Hofacker. ViennaRNA Package 2.0. In: *Algorithms for Molecular Biology* 6.1 (2011), p. 26. DOI: 10.1186/1748-7188-6-26.

[58] J. Abramson, J. Adler, J. Dunger, et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. In: *Nature* 630.8016 (2024), pp. 493–500. DOI: 10.1038/s41586-024-07487-w.

Overview of AI Technologies Used

ChatGPT (OpenAI) was used as a supporting tool for language refinement, formatting, and code troubleshooting. All scientific analyses, interpretations and arguments were produced by the author.